

REGULARITY OF MINIMAL SURFACES AND THE BERNSTEIN PROBLEM

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ABSTRACT. These notes are based on lectures given by Nick Edelen in Xiamen and taken by Wei Wang. The lectures focused on the regularity theory of minimal surfaces through the language of varifolds, currents, finite-perimeter sets, monotonicity, tangent cones, Allard regularity, and dimension reduction. Because of time constraints, the Bernstein theorem was not covered in the course; a short final chapter on Bernstein-type results has been added, mainly following the presentation in Colding–Minicozzi’s book.

1. INTRODUCTION

1.1. Motivation Question: Plateau Problem. The motivating variational question is the following form of the Plateau problem. Let $\Gamma^{n-1} \subset \mathbb{R}^{n+m}$ be a prescribed boundary. Does there exist an n -dimensional surface $M^n \subset \mathbb{R}^{n+m}$ such that $\partial M = \Gamma$ and

$$\mathcal{H}^n(M) = \inf\{\mathcal{H}^n(N) : N^n \subset \mathbb{R}^{n+m}, \partial N = \Gamma\}?$$

Here $\mathcal{H}^n(M)$ denotes the n -dimensional area of M . Physically, one should think of a soap film spanning a fixed wire frame Γ : surface tension drives the film toward smaller area while the boundary is held fixed. Soap bubbles lead to a related constrained problem, where one also fixes an enclosed volume and obtains constant mean curvature rather than zero mean curvature. In the unconstrained fixed-boundary problem, the ideal picture is that the area functional has a minimizer among all admissible spanning surfaces.

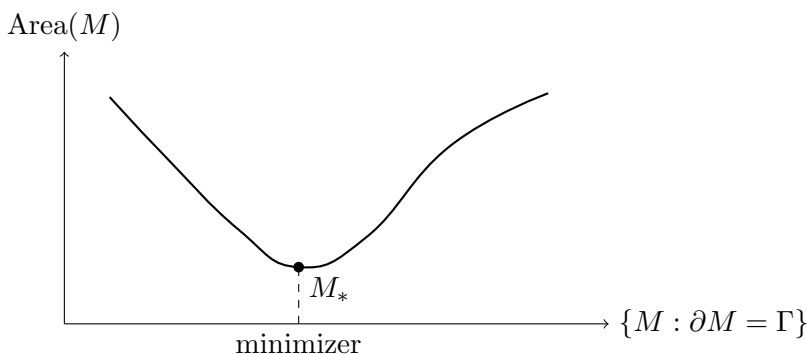


FIGURE 1. A schematic view of the Plateau problem: among all surfaces with boundary Γ , one seeks a surface M_* with least area. The horizontal axis is only a mnemonic for the class of admissible surfaces.

This simple formulation hides two central mathematical difficulties. First, a minimizing sequence of smooth surfaces need not converge to a smooth surface in any classical sense, so one must choose a weak category, such as currents or varifolds, in which compactness is available. Second, even after an object minimizing area is constructed, its regularity is a separate problem. The Bernstein problem is another expression of the same theme: it asks when an entire minimal graph must be flat, and hence connects the equation $H = 0$ with global rigidity.

1.2. Minimal Surfaces and the First Variation. We now pass from global minimization to the infinitesimal condition satisfied by a smooth minimizer. Let $M^n \subset \mathbb{R}^{n+m}$ be a smooth submanifold, possibly with boundary. A variation of M is a smooth family of diffeomorphisms $\varphi_t : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^{n+m}$, defined for $|t|$ small, such that $\varphi_0 = \text{id}$. Its variational vector field is

$$X(x) = \left. \frac{d}{dt} \right|_{t=0} \varphi_t(x).$$

If we want to keep the boundary $\Gamma = \partial M$ fixed, we require $\varphi_t = \text{id}$ on Γ , equivalently $X = 0$ on Γ . We write $M_t = \varphi_t(M)$. A smooth surface M with fixed boundary is called stationary, or minimal in the smooth sense, if

$$\left. \frac{d}{dt} \right|_{t=0} \mathcal{H}^n(M_t) = 0$$

for every such boundary-preserving variation. An area-minimizing surface is stationary whenever the first variation is well defined, but stationarity is weaker than being a global minimizer.

Proposition 1.1 (First variation formula). *Let $M^n \subset \mathbb{R}^{n+m}$ be a smooth compact submanifold with boundary, and let φ_t be a variation with variational vector field X . Then*

$$\left. \frac{d}{dt} \right|_{t=0} \mathcal{H}^n(\varphi_t(M)) = \int_M \text{div}_M X \, d\mathcal{H}^n.$$

If $e_1, \dots, e_n$ is a local orthonormal frame for TM , the mean curvature vector is

$$H_M = \sum_{i=1}^n (D_{e_i} e_i)^\perp,$$

where D is the Euclidean connection and $(\cdot)^\perp$ denotes projection onto the normal bundle of M . With this convention,

$$\left. \frac{d}{dt} \right|_{t=0} \mathcal{H}^n(\varphi_t(M)) = \int_{\partial M} \langle X, \eta \rangle \, d\mathcal{H}^{n-1} - \int_M \langle H_M, X \rangle \, d\mathcal{H}^n,$$

where η is the outward unit conormal along ∂M . In particular, if $X = 0$ on ∂M , then

$$\left. \frac{d}{dt} \right|_{t=0} \mathcal{H}^n(\varphi_t(M)) = - \int_M \langle H_M, X \rangle \, d\mathcal{H}^n.$$

Proof. Fix $x \in M$, and choose an orthonormal basis $e_1, \dots, e_n$ of $T_x M$. The area formula gives

$$\mathcal{H}^n(\varphi_t(M)) = \int_M J_M \varphi_t \, d\mathcal{H}^n, \quad J_M \varphi_t = \sqrt{\det(g_{ij}(t))},$$

where $g_{ij}(t) = \langle D\varphi_t(e_i), D\varphi_t(e_j) \rangle$. Since $D\varphi_t = I + tDX + o(t)$, we have

$$g_{ij}(t) = \delta_{ij} + t\{\langle D_{e_i} X, e_j \rangle + \langle e_i, D_{e_j} X \rangle\} + o(t).$$

Using $\det(I + tA + o(t)) = 1 + t \, \text{tr} A + o(t)$, it follows that

$$\left. \frac{d}{dt} \right|_{t=0} J_M \varphi_t = \frac{1}{2} \text{tr}(g'(0)) = \sum_{i=1}^n \langle D_{e_i} X, e_i \rangle = \text{div}_M X.$$

Integrating over M proves the first identity. To identify the geometric meaning of this expression, decompose $X = X^T + X^\perp$ into its tangential and normal parts. For the normal part,

$$\text{div}_M X^\perp = \sum_{i=1}^n \langle D_{e_i} X^\perp, e_i \rangle = - \sum_{i=1}^n \langle X^\perp, D_{e_i} e_i \rangle = - \langle H_M, X \rangle.$$

For the tangential part, the divergence theorem on M gives

$$\int_M \text{div}_M X^T \, d\mathcal{H}^n = \int_{\partial M} \langle X, \eta \rangle \, d\mathcal{H}^{n-1}.$$

Combining these two identities yields the stated formula. $\square$

The formula immediately gives the pointwise equation for smooth minimal surfaces. If $X = 0$ on ∂M , then stationarity for every X implies $H_M \equiv 0$, because the normal component of X can be chosen freely in the interior. Conversely, if $H_M \equiv 0$, the first variation vanishes for every boundary-preserving variation. Thus

$$M \text{ is minimal} \iff H_M \equiv 0.$$

There is also a useful analytic form of the same condition. Let $x : M \rightarrow \mathbb{R}^{n+m}$ be the inclusion map. The vector-valued Laplace–Beltrami operator satisfies

$$\Delta_M x = \sum_{i=1}^n \{D_{e_i}(Dx(e_i)) - Dx(\nabla_{e_i}^M e_i)\} = \sum_{i=1}^n (D_{e_i} e_i - \nabla_{e_i}^M e_i) = H_M.$$

Therefore

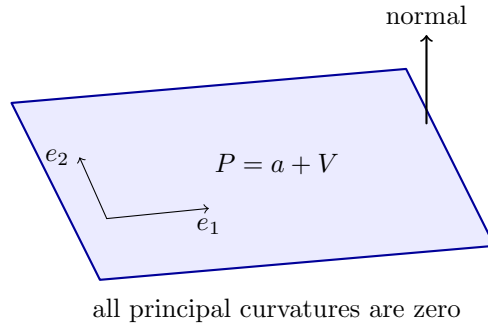
$$M \text{ is minimal} \iff H_M \equiv 0 \iff \Delta_M x = 0.$$

The last condition means that every ambient coordinate function $x^\alpha|_M$, $1 \leq \alpha \leq n+m$, is harmonic with respect to the Laplace–Beltrami operator on M .

1.3. Basic Examples.

We now record several standard examples. The pictures below are schematic; for the higher-dimensional examples they show the symmetry reduction or the singular pattern rather than a literal embedding in the page.

(1) Planes. Every affine plane $P = a + V \subset \mathbb{R}^{n+m}$ is minimal. Indeed, its second fundamental form vanishes identically, so $H_P = 0$.

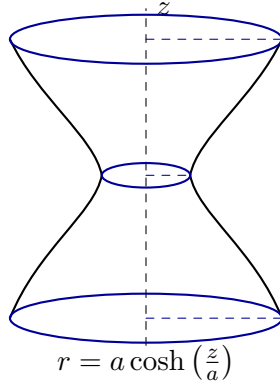


(2) Catenoid. The catenoid is the surface obtained by rotating the catenary $r = a \cosh\left(\frac{z}{a}\right)$ around the z -axis:

$$X(z, \theta) = \left(a \cosh\left(\frac{z}{a}\right) \cos \theta, a \cosh\left(\frac{z}{a}\right) \sin \theta, z \right).$$

It is the basic non-flat minimal surface of revolution.

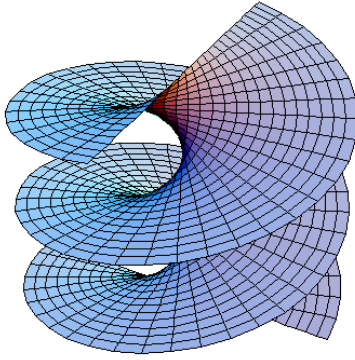
catenary rotated about the axis



(3) Helicoid. The helicoid is parametrized by

$$X(s, t) = (s \cos t, s \sin t, at),$$

and is swept out by a line that rotates while translating along its axis. Together with the plane, it is a basic ruled minimal surface.

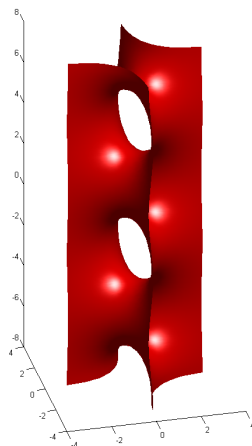


Helicoid: a rotating family of straight rulings.

(4) Scherk surfaces. Scherk's surfaces are the next classical examples after the catenoid and helicoid. One model is the graph

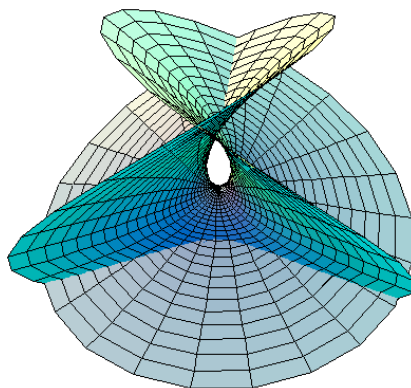
$$z = \log \frac{\cos y}{\cos x},$$

on a square, and the singly periodic Scherk surface may be viewed geometrically as a minimal desingularization of two intersecting planes.



Singly periodic Scherk surface.

(5) **Enneper surface.** Enneper's surface is a complete immersed minimal surface with self-intersections. It is produced by the Weierstrass data $g(z) = z$ and $dh = dz$, and it is useful as a simple non-embedded example.



Enneper surface from the data $g(z) = z$, $dh = dz$.

(6) **Costa–Hoffman–Meeks examples.** The Costa surface is a complete embedded minimal surface of genus one with three ends. Hoffman and Meeks extended this picture to embedded examples of higher genus; the Costa–Hoffman–Meeks family has surfaces of different genus, still with three ends. The image shows the genus-one case.



Costa surface, the genus-one member of the Costa–Hoffman–Meeks picture.

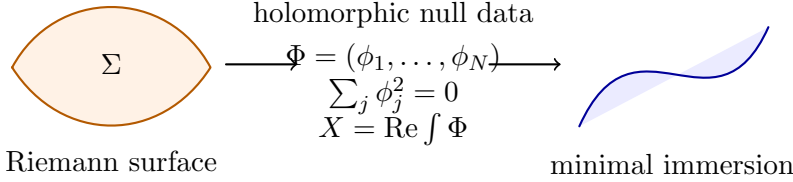
(7) Weierstrass representation. Let $X : \Sigma \rightarrow \mathbb{R}^N$ be a conformal immersion from a Riemann surface. In a conformal coordinate $z = u + iv$, the induced metric is $\lambda^2(du^2 + dv^2)$, so

$$\Delta_M X = \lambda^{-2}(X_{uu} + X_{vv}).$$

Thus X is minimal if and only if its coordinate functions are harmonic. Hence $\phi_j = \partial_z X^j dz$ are holomorphic one-forms, and conformality gives the null condition $\sum_{j=1}^N \phi_j^2 = 0$. Conversely, holomorphic one-forms $\phi_1, \dots, \phi_N$ satisfying $\sum_j \phi_j^2 = 0$ and $\sum_j |\phi_j|^2 > 0$ define locally

$$X = \operatorname{Re} \int (\phi_1, \dots, \phi_N),$$

which is conformal and harmonic, hence minimal by $\Delta_M X = 0$.



(8) Minimal graphs and the minimal surface system. Let $u : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be a C^2 scalar function and let $M = \text{graph } u$. Then

$$g_{ij} = \delta_{ij} + D_i u D_j u, \quad g^{ij} = \delta_{ij} - \frac{D_i u D_j u}{1 + |Du|^2}.$$

The graph is minimal if and only if u satisfies the scalar minimal surface equation

$$\operatorname{div} \left(\frac{Du}{\sqrt{1 + |Du|^2}} \right) = 0,$$

equivalently $\sum_{i,j=1}^n g^{ij} D_{ij} u = 0$. If u is a Lipschitz weak solution of this scalar equation, then the gradient bound makes the equation uniformly elliptic; standard elliptic regularity and bootstrapping imply that u is smooth. For a vector-valued map $u : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}^m$, the induced metric is

$$g_{ij} = \delta_{ij} + \sum_{\alpha=1}^m D_i u^\alpha D_j u^\alpha,$$

and the minimal graph equation becomes the minimal surface system

$$\sum_{i,j=1}^n g^{ij} D_{ij} u^\alpha = 0, \quad \alpha = 1, \dots, m.$$

In higher codimension the scalar regularity statement fails: the Lawson–Osserman example

$$u(x) = \frac{\sqrt{5}}{2} |x| \sigma \left(\frac{x}{|x|} \right), \quad x \in \mathbb{R}^4,$$

where $\sigma : \mathbb{S}^3 \rightarrow \mathbb{S}^2 \subset \mathbb{R}^3$ is the Hopf fibration, is Lipschitz and one-homogeneous, but is not differentiable at the origin; its graph is a stationary minimal cone in $\mathbb{R}^7$.

(9) Holomorphic subvarieties. Let (X, J, g, ω) be a Kähler manifold, and let $V \subset X$ be a complex k -dimensional submanifold, or more generally a holomorphic subvariety viewed as an integer multiplicity rectifiable current. The form

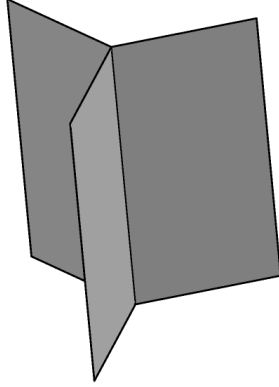
$$\phi = \frac{\omega^k}{k!}$$

is closed and has comass at most one by Wirtinger's inequality, and on V_{reg} it equals the Riemannian volume form. Therefore, if V' is homologous to V , then

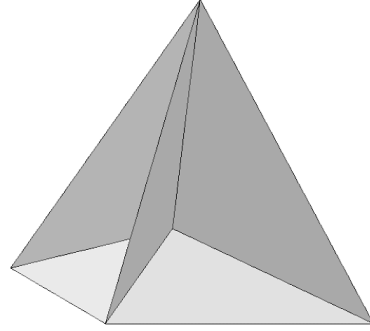
$$\text{Vol}(V) = \int_V \phi = \int_{V'} \phi \leq \text{Vol}(V').$$

Thus holomorphic subvarieties are area-minimizing in their homology class, and in particular their smooth parts are minimal.

(10) The Y and T cones. These are not smooth surfaces, but they are basic minimal cones for soap-film type problems in $\mathbb{R}^3$. The Y cone is the product of a line with three half-lines meeting at angles 120° . The T cone is the cone over the one-skeleton of a regular tetrahedron; at the vertex four triple junction curves meet with the tetrahedral angle $\arccos(-\frac{1}{3})$.



Y cone



T cone

(11) Simons cone. The Simons cone is

$$C = \{(x, y) \in \mathbb{R}^4 \times \mathbb{R}^4 : |x| = |y|\} \subset \mathbb{R}^8.$$

It is a minimal hypersurface cone with an isolated singularity at the origin. It is area-minimizing in $\mathbb{R}^8$ and is one of the model examples behind the change in Bernstein-type regularity phenomena in high dimension.

2. RECTIFIABLE SETS AND VARIFOLDS

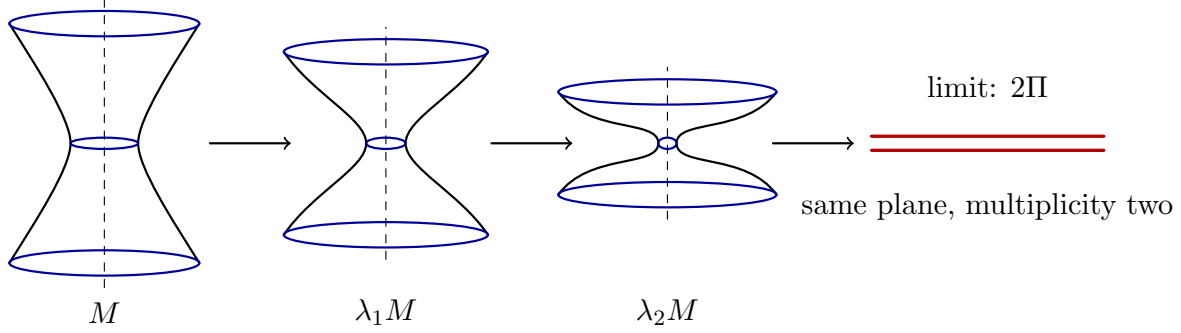
2.1. Motivation: Weak Notions of Surfaces. The next step is to enlarge the class of admissible surfaces. Smooth submanifolds are too rigid for several basic operations in the calculus of variations: limits may develop singularities, multiplicities, or concentration of area. The goal is to introduce weak notions of surfaces, first rectifiable sets and then varifolds, which still remember area, tangent planes, and first variation.

There are two main motivations.

- (1) *Allow singularities.* Minimal objects produced by variational problems need not be smooth everywhere; soap films already show triple junctions and cone-like points. A weak surface should allow such behavior while still having an area and an infinitesimal tangent plane almost everywhere.
- (2) *Recover compactness.* We want a topology in which bounded area gives subsequential convergence. This includes the direct method for the Plateau problem: choose a minimizing sequence M_i with $\partial M_i = \Gamma$, pass to a limit, and prove that the limit still spans Γ and minimizes area. The analogy is the passage from smooth functions to Sobolev functions: weaken the objects enough to obtain compactness, but keep enough structure to write the Euler–Lagrange equation.

A first compactness picture already appears in the catenoid. If M is a catenoid and $\lambda_i \downarrow 0$, then the blow-downs $\lambda_i M$ flatten out; in the varifold sense they converge to the horizontal plane with

multiplicity two. The point is not merely that the picture becomes flatter, but that the limiting object must remember how much area arrives at the plane.



The direct method is therefore part of the compactness problem. If M_i is a minimizing sequence with $\partial M_i = \Gamma$, a naive smooth or pointwise notion of convergence is not adequate:

- (1) Thin necks and tubes may appear and then disappear in the set-theoretic limit, even though they carry area along the sequence.
- (2) Multiplicity can be lost if the limit only records the underlying subset.
- (3) The first variation must pass to the limit, since the Euler–Lagrange equation is the geometric content of minimality.

Varifolds are designed to address exactly these points: they give a weak topology in which bounded mass has compactness and the first variation still has a distributional meaning.

2.2. Hausdorff Measure and Rectifiable Sets.

Definition 2.1. Let $E \subset \mathbb{R}^N$, where in our applications $N = n + m$. For $s \geq 0$ and $\delta > 0$ define

$$\mathcal{H}_\delta^s(E) = \inf \left\{ \sum_j \alpha_s \left(\frac{\text{diam } U_j}{2} \right)^s : E \subset \bigcup_j U_j, \text{diam } U_j < \delta \right\},$$

where

$$\alpha_s = \frac{\pi^{\frac{s}{2}}}{\Gamma(1 + \frac{s}{2})}$$

is the volume of the unit ball in $\mathbb{R}^s$ when s is an integer. The s -dimensional Hausdorff measure is

$$\mathcal{H}^s(E) = \lim_{\delta \downarrow 0} \mathcal{H}_\delta^s(E) = \sup_{\delta > 0} \mathcal{H}_\delta^s(E),$$

and the Hausdorff dimension is

$$\dim_{\mathcal{H}} E = \inf \{s : \mathcal{H}^s(E) = 0\} = \sup \{s : \mathcal{H}^s(E) = \infty\}.$$

Basic properties. This normalization gives $\mathcal{H}^N = \mathcal{L}^N$ on $\mathbb{R}^N$, and if $M^k \subset \mathbb{R}^N$ is a smooth k -dimensional submanifold, then $\mathcal{H}^k \llcorner M$ is the usual Riemannian area measure on M . We shall use the following elementary facts repeatedly.

- (1) $\mathcal{H}^s$ is a Borel regular outer measure, monotone and countably subadditive.
- (2) If $\mathcal{H}^s(E) < \infty$ and $t > s$, then $\mathcal{H}^t(E) = 0$; if $\mathcal{H}^s(E) > 0$ and $t < s$, then $\mathcal{H}^t(E) = \infty$. Thus the Hausdorff dimension is the critical exponent where the measure jumps from ∞ to 0.
- (3) Countable sets have $\mathcal{H}^s$ -measure zero for every $s > 0$, while every nonempty open set in $\mathbb{R}^N$ has Hausdorff dimension N .
- (4) If $f : E \rightarrow \mathbb{R}^L$ is L -Lipschitz, then

$$\mathcal{H}^s(f(E)) \leq L^s \mathcal{H}^s(E),$$

so Lipschitz maps do not increase Hausdorff dimension. In particular, orthogonal projections do not increase $\mathcal{H}^s$.

Definition 2.2. A set $M \subset \mathbb{R}^N$ is called countably k -rectifiable if there are Lipschitz maps $F_j : \mathbb{R}^k \rightarrow \mathbb{R}^N$ and a set M_0 with $\mathcal{H}^k(M_0) = 0$ such that

$$M \subset M_0 \cup \bigcup_{j=1}^{\infty} F_j(\mathbb{R}^k).$$

If in addition $\mathcal{H}^k(M) < \infty$ (or locally finite on compact subsets), we often simply call M a k -rectifiable set.

Remark 2.3 (Lipschitz charts versus C^1 charts). Equivalently, up to an $\mathcal{H}^k$ -null set, one may cover M by countably many Lipschitz graphs, or by countably many C^1 graphs. The implication from C^1 to Lipschitz is local and elementary.

Conversely, for a Lipschitz graph, Rademacher's theorem gives differentiability a.e. By Lusin's theorem the derivative is continuous on large compact pieces, and Whitney's extension theorem upgrades those pieces to C^1 graphs. This is the sense in which Lipschitz and C^1 graphs are interchangeable in the definition.

Example 2.4 (Smooth and ugly rectifiable sets). Every C^1 k -submanifold is countably k -rectifiable. A much uglier example is obtained as follows: enumerate $\mathbb{Q}^N = \{q_i\}_{i=1}^{\infty}$, fix $1 \leq k \leq N-1$, and set

$$\mathbb{S}_r^k = \{x \in \mathbb{R}^{k+1} : |x| = r\} \times \{0\}^{N-k-1} \subset \mathbb{R}^N, \quad M = \bigcup_{i=1}^{\infty} (q_i + \mathbb{S}_{2^{-i}}^k).$$

For the n -dimensional theory below one takes $k = n$. This M is a countable union of smooth k -spheres, hence countably k -rectifiable, and for $k \geq 1$ it has finite $\mathcal{H}^k$ -measure because $\sum_i 2^{-ik} < \infty$. Nevertheless it is dense in $\mathbb{R}^N$ and has no reasonable manifold-like local appearance. Rectifiability is therefore a measure-theoretic notion, not a topological smoothness condition.

Definition 2.5. Let M be $\mathcal{H}^k$ -measurable with locally finite $\mathcal{H}^k$ -measure. We say that a k -plane $T \in G(N, k)$ is the approximate tangent plane to M at x , written $T_x M = T$, if the blown-up measures converge to the flat measure on T :

$$\lim_{\rho \downarrow 0} \rho^{-k} \int_M \phi \left(\frac{y-x}{\rho} \right) d\mathcal{H}^k(y) = \int_T \phi(z) d\mathcal{H}^k(z), \quad \phi \in C_c(\mathbb{R}^N).$$

Equivalently, after rescaling around x , the set M becomes indistinguishable in measure from the plane T .

Proposition 2.6 (Approximate tangent planes exist a.e.). *If $M \subset \mathbb{R}^N$ is countably k -rectifiable and locally $\mathcal{H}^k$ -finite, then $T_x M$ exists for $\mathcal{H}^k$ -a.e. $x \in M$.*

Proof. Cover M up to an $\mathcal{H}^k$ -null set by countably many C^1 graphs Γ_j . Replace this cover by a disjoint measurable decomposition

$$E_j \subset M \cap \Gamma_j,$$

still covering M up to a null set. By the differentiation theorem on the C^1 manifold Γ_j , for $\mathcal{H}^k$ -a.e. $x \in E_j$ the set E_j has density one inside Γ_j .

Since a C^1 graph has tangent plane $T_x \Gamma_j$ and its blow-ups converge to $T_x \Gamma_j$, the density-one property implies that the blow-ups of E_j , and hence of M , have the same limit. Thus $T_x M = T_x \Gamma_j$ for $\mathcal{H}^k$ -a.e. $x \in E_j$. Taking the countable union over j gives the result. $\square$

Definition 2.7. Let M be countably k -rectifiable and let $f : M \rightarrow \mathbb{R}^\ell$ be Lipschitz. On each rectifiable chart, Rademacher's theorem gives a derivative a.e.; these derivatives agree on overlaps and define the tangential differential

$$d^M f_x : T_x M \rightarrow \mathbb{R}^\ell, \quad \mathcal{H}^k\text{-a.e. } x \in M.$$

For scalar f we write $\nabla^M f$ for the vector in $T_x M$ representing $d^M f_x$. If $\ell \geq k$, the k -dimensional tangential Jacobian is

$$J_M f(x) = \sqrt{\det((d^M f_x)^* d^M f_x)}.$$

If $\ell \leq k$, the coarea Jacobian is

$$J_\ell^M f(x) = \sqrt{\det(d^M f_x (d^M f_x)^*)}.$$

Theorem 2.8 (Area formula). *If $A \subset M$ is $\mathcal{H}^k$ -measurable, M is countably k -rectifiable, and $f : M \rightarrow \mathbb{R}^\ell$ is Lipschitz with $\ell \geq k$, then*

$$\int_A J_M f \, d\mathcal{H}^k = \int_{\mathbb{R}^\ell} N(f, A, y) \, d\mathcal{H}^k(y), \quad N(f, A, y) = \mathcal{H}^0(A \cap f^{-1}\{y\}).$$

In particular $f(A)$ is countably k -rectifiable. If f is one-to-one on A , then

$$\mathcal{H}^k(f(A)) = \int_A J_M f \, d\mathcal{H}^k.$$

If f is not one-to-one, the same formula counts multiplicity: points y with several preimages contribute several times, so only the inequality $\mathcal{H}^k(f(A)) \leq \int_A J_M f \, d\mathcal{H}^k$ remains in general.

Theorem 2.9 (Coarea formula). *If $A \subset M$ is $\mathcal{H}^k$ -measurable, M is countably k -rectifiable, and $f : M \rightarrow \mathbb{R}^\ell$ is Lipschitz with $\ell \leq k$, then*

$$\int_A J_\ell^M f \, d\mathcal{H}^k = \int_{\mathbb{R}^\ell} \mathcal{H}^{k-\ell}(A \cap f^{-1}\{y\}) \, d\mathcal{L}^\ell(y).$$

Moreover, for $\mathcal{L}^\ell$ -a.e. y , the slice $A \cap f^{-1}\{y\}$ is countably $(k - \ell)$ -rectifiable. Thus the area formula controls images of rectifiable sets, while the coarea formula controls almost every level set.

2.3. Integral Varifolds, Weak Convergence, and First Variation.

Definition 2.10. Let $U \subset \mathbb{R}^N$ be open. Denote by $G(N, n)$ the Grassmannian of unoriented n -planes in $\mathbb{R}^N$, and set

$$G_n(U) = U \times G(N, n).$$

An n -varifold in U is a Radon measure on $G_n(U)$. This definition records both position and tangent-plane information. If $M \subset U$ is countably n -rectifiable and $\theta : M \rightarrow [0, \infty)$ is locally $\mathcal{H}^n$ -integrable, then the associated rectifiable varifold is

$$V = \underline{v}(M, \theta), \quad V(\phi) = \int_M \phi(x, T_x M) \theta(x) \, d\mathcal{H}^n(x), \quad \phi \in C_c(G_n(U)).$$

The function θ is the multiplicity or density. If $\theta(x) \in \mathbb{N}$ for $\mathcal{H}^n$ -a.e. x , then V is an integral varifold. If $\theta \equiv 1$, we write

$$|M| = \underline{v}(M, 1),$$

the multiplicity-one varifold carried by M .

Remark 2.11 (Two equivalent viewpoints). The Grassmannian definition and the rectifiable notation are the same object viewed from two angles. A general varifold is any Radon measure V on $G_n(U)$; a rectifiable varifold is the special case where this measure is concentrated on the graph $x \mapsto T_x M$ and is weighted by $\theta \mathcal{H}^n$.

Definition 2.12. The mass, or weight, measure of V is the Radon measure $\|V\|$ on U defined by

$$\|V\|(A) = V(A \times G(N, n)), \quad A \subset U.$$

For $V = \underline{v}(M, \theta)$ this becomes

$$\|V\|(A) = \int_{M \cap A} \theta \, d\mathcal{H}^n.$$

The total mass in $W \subset U$ is $\|V\|(W)$.

Definition 2.13. Weak convergence of varifolds is weak convergence as Radon measures on $G_n(U)$:

$$V_i \rightharpoonup V \iff \int_{G_n(U)} \phi dV_i \rightarrow \int_{G_n(U)} \phi dV \quad \text{for every } \phi \in C_c(G_n(U)).$$

Equivalently, for rectifiable varifolds $V_i = \underline{v}(M_i, \theta_i)$,

$$\int_{M_i} \phi(x, T_x M_i) \theta_i d\mathcal{H}^n \rightarrow \int_M \phi(x, T_x M) \theta d\mathcal{H}^n.$$

Taking test functions independent of the plane variable gives $\|V_i\| \xrightarrow{*} \|V\|$ as Radon measures, but the converse is false: the tangent planes may oscillate even when the mass measures converge.

Example 2.14 (Oscillating sawtooth varifolds). In the xy -plane take the isosceles triangle with vertices

$$O = (0, 0), \quad A = (b, a), \quad B = (2b, 0), \quad a, b > 0.$$

Let M_1 be the two sides $OA \cup AB$. Reflect the top triangle across the midline to obtain two smaller tents; repeating this gives a sawtooth curve M_i with vertices

$$p_j^{(i)} = \left(\frac{bj}{2^{i-1}}, \begin{cases} 0, & j \text{ even,} \\ \frac{a}{2^{i-1}}, & j \text{ odd,} \end{cases} \right), \quad j = 0, \dots, 2^i.$$

Thus M_i has 2^i segments, each of length $\frac{\sqrt{a^2+b^2}}{2^{i-1}}$, and each segment has density one:

$$|M_i| = \underline{v}(M_i, 1), \quad \||M_i|\|(A) = \mathcal{H}^1(M_i \cap A),$$

$$\||M_i|\|(\mathbb{R}^2) = \mathcal{H}^1(M_i) = 2^i \frac{\sqrt{a^2+b^2}}{2^{i-1}} = 2\sqrt{a^2+b^2}.$$

Let $I = [0, 2b] \times \{0\}$ and let

$$S_+ = \text{span}(b, a), \quad S_- = \text{span}(b, -a), \quad c = \frac{\sqrt{a^2+b^2}}{b}.$$

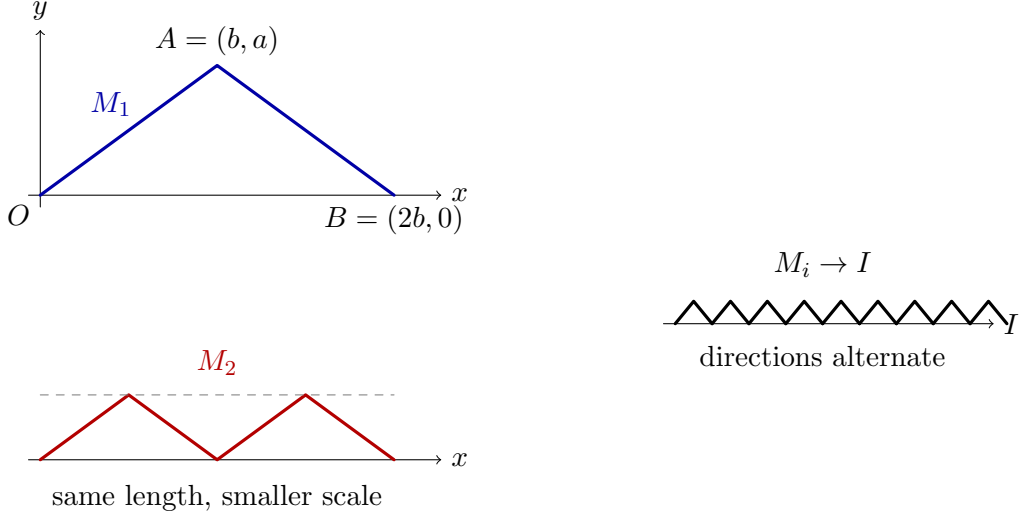
Then M_i converges to I as a set, while the varifolds converge to the non-integral-looking limit

$$V(\phi) = \frac{c}{2} \int_0^{2b} [\phi((x, 0), S_+) + \phi((x, 0), S_-)] dx.$$

Consequently

$$\||M_i|\| \xrightarrow{*} c\mathcal{H}^1 \llcorner I, \quad \|V\| = c\mathcal{H}^1 \llcorner I.$$

The limiting mass remembers the excess length of the oscillating sawtooth, and the limiting varifold remembers that the tangent directions oscillated between S_+ and S_- rather than becoming horizontal.



Definition 2.15. Pushforward is the operation that makes deformations precise. Let $f : U \rightarrow U'$ be C^1 and suppose Df_x has rank n on S for V -a.e. (x, S) where the Jacobian below is nonzero. Define

$$f_{\#}V(\phi) = \int_{G_n(U)} \phi(f(x), Df_x(S)) J_S f(x) dV(x, S),$$

where $J_S f(x) = \sqrt{\det((Df_x|_S)^*(Df_x|_S))}$. For $V = \underline{v}(M, \theta)$ this agrees with the area formula: if f is one-to-one on M , then $f_{\#}V = \underline{v}(f(M), \theta \circ f^{-1})$; if not, multiplicities from different preimages add.

The corresponding mass measure is obtained by testing only the position variable:

$$\|f_{\#}V\|(A) = \int_{G_n(U)} \mathbf{1}_A(f(x)) J_S f(x) dV(x, S), \quad A \subset U' \text{ Borel.}$$

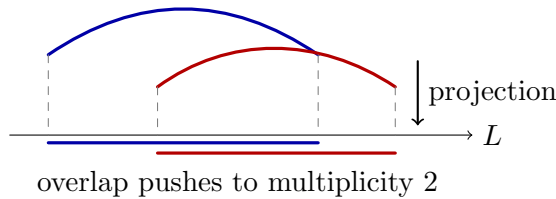
Thus

$$\|f_{\#}V\|(U') = \int_{G_n(U)} J_S f(x) dV(x, S).$$

In the rectifiable case this becomes

$$\|f_{\#}\underline{v}(M, \theta)\|(A) = \int_{M \cap f^{-1}(A)} J_M f(x) \theta(x) d\mathcal{H}^n(x).$$

When f is not one-to-one, this formula is exactly where the multiplicities of the different sheets are added.



Definition 2.16. Let $X \in C_c^1(U; \mathbb{R}^N)$ and let Φ_t be its flow, $\frac{d}{dt}\Phi_t(x) = X(\Phi_t(x))$, $\Phi_0 = \text{id}$. The first variation of V is

$$\delta V(X) = \left. \frac{d}{dt} \right|_{t=0} \|(\Phi_t)_{\#}V\|(U).$$

Using the pushforward formula and differentiating the tangential Jacobian gives

$$\delta V(X) = \int_{G_n(U)} \text{div}_S X(x) dV(x, S),$$

where, if $e_1, \dots, e_n$ is an orthonormal basis of S ,

$$\operatorname{div}_S X = \sum_{i=1}^n e_i \cdot D_{e_i} X = \operatorname{tr}(P_S DX).$$

For a rectifiable varifold this reads

$$\delta \underline{v}(M, \theta)(X) = \int_M \theta \operatorname{div}_M X d\mathcal{H}^n = \int_U \operatorname{div}_{T_x M} X d\|\underline{v}(M, \theta)\|(x).$$

We call V stationary, or minimal in the weak varifold sense, if $\delta V(X) = 0$ for every $X \in C_c^1(U; \mathbb{R}^N)$. When M is a smooth multiplicity-one submanifold without boundary in U , this reduces to the classical condition $H_M = 0$.

Example 2.17 (Smooth stationary submanifolds). Let $M^n \subset U \subset \mathbb{R}^{n+m}$ be a smooth submanifold with smooth boundary ∂M , and let $|M| = \underline{v}(M, 1)$. For $X \in C_c^1(U; \mathbb{R}^{n+m})$,

$$\delta |M|(X) = \int_M \operatorname{div}_M X d\mathcal{H}^n.$$

Decompose $X = X^T + X^\perp$ along M . With the sign convention used above,

$$\operatorname{div}_M X = \operatorname{div}_M X^T - H_M \cdot X,$$

and the divergence theorem on the smooth manifold M gives

$$\delta |M|(X) = \int_{\partial M} X \cdot \eta d\mathcal{H}^{n-1} - \int_M H_M \cdot X d\mathcal{H}^n,$$

where η is the outward unit conormal to ∂M in M . Thus, if M is smooth and stationary in the interior, equivalently $H_M = 0$, then the first variation reduces to the boundary term

$$\delta |M|(X) = \int_{\partial M} X \cdot \eta d\mathcal{H}^{n-1}.$$

In particular, if $\partial M = \emptyset$ in U , or if X vanishes on ∂M , then $\delta |M|(X) = 0$.

Example 2.18 (Full-dimensional stationary integral varifolds). Let V be an n -dimensional integral varifold in $\mathbb{R}^n$. Since $G(n, n)$ consists of the single plane $\mathbb{R}^n$, the varifold is completely determined by its mass measure. By integrality there is an integer-valued function $\Theta \in L_{\text{loc}}^1(\mathbb{R}^n; \mathbb{N} \cup \{0\})$ such that

$$\|V\| = \Theta \mathcal{L}^n, \quad V = \underline{v}(\mathbb{R}^n, \Theta).$$

Indeed, one may take $\Theta = \theta \mathbf{1}_M$ if $V = \underline{v}(M, \theta)$, because $\mathcal{H}^n = \mathcal{L}^n$ in $\mathbb{R}^n$ and the approximate tangent plane is $\mathbb{R}^n$ at $\mathcal{L}^n$ -a.e. density point of M .

Assume now that V is stationary. For every $X \in C_c^1(\mathbb{R}^n; \mathbb{R}^n)$,

$$0 = \delta V(X) = \int_{\mathbb{R}^n} \operatorname{div} X d\|V\| = \int_{\mathbb{R}^n} \Theta \operatorname{div} X dx.$$

Taking $X = \phi e_j$ with $\phi \in C_c^1(\mathbb{R}^n)$ gives

$$\int_{\mathbb{R}^n} \Theta \partial_j \phi dx = 0, \quad j = 1, \dots, n.$$

Hence the distributional gradient $D\Theta$ is zero. To see what this means concretely, mollify: for any standard mollifier ρ_ε , the smooth function $\Theta_\varepsilon = \rho_\varepsilon * \Theta$ satisfies $\nabla \Theta_\varepsilon = 0$ on every ball where the convolution is defined, so Θ_ε is locally constant. Since $\mathbb{R}^n$ is connected, the constants agree on overlaps; letting $\varepsilon \downarrow 0$ gives $\Theta \equiv q$ a.e. for some $q \in \mathbb{N} \cup \{0\}$. Therefore

$$V = q |\mathbb{R}^n|.$$

Conversely, every constant-multiplicity plane $q|\mathbb{R}^n|$ is stationary, since

$$\delta(q|\mathbb{R}^n|)(X) = q \int_{\mathbb{R}^n} \operatorname{div} X \, dx = 0$$

for compactly supported X . Thus the only full-dimensional integral stationary varifolds in $\mathbb{R}^n$ are constant integer multiples of the whole space.

Example 2.19 (Stationary Y -junction). Let $\tau_1, \tau_2, \tau_3 \in \mathbb{R}^2$ be unit vectors with mutual angles 120° , and let

$$R_i = \{t\tau_i : t \geq 0\}, \quad V = |R_1| + |R_2| + |R_3|.$$

Here each ray has density one. Since $T_x R_i = \operatorname{span}(\tau_i)$ away from the origin, for $X \in C_c^1(\mathbb{R}^2; \mathbb{R}^2)$,

$$\delta V(X) = \sum_{i=1}^3 \int_{R_i} \operatorname{div}_{R_i} X \, d\mathcal{H}^1 = \sum_{i=1}^3 \int_0^\infty \tau_i \cdot DX(t\tau_i) \tau_i \, dt.$$

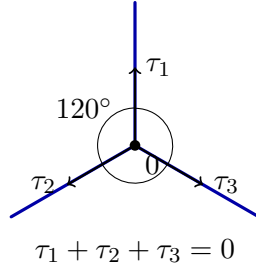
But

$$\tau_i \cdot DX(t\tau_i) \tau_i = \frac{d}{dt} (X(t\tau_i) \cdot \tau_i),$$

so the one-dimensional divergence theorem gives

$$\delta V(X) = \sum_{i=1}^3 [X(t\tau_i) \cdot \tau_i]_{t=0}^\infty = -X(0) \cdot (\tau_1 + \tau_2 + \tau_3).$$

For three unit vectors meeting at 120° one has $\tau_1 + \tau_2 + \tau_3 = 0$, and therefore $\delta V(X) = 0$ for every compactly supported X . Thus the Y -junction is a stationary integral varifold, even though it is not a smooth submanifold at the origin.



Remark 2.20 (Balancing condition for line segments). The same computation works for any finite union of straight line segments or rays with integer densities θ_e . At an interior junction point p , let $\tau_{e,p}$ be the unit tangent vector along the edge e pointing away from p . The first variation has a boundary contribution

$$-X(p) \cdot \sum_{e \ni p} \theta_e \tau_{e,p}.$$

Hence stationarity at every free junction is equivalent to the balancing condition

$$\sum_{e \ni p} \theta_e \tau_{e,p} = 0$$

at each such point. For three density-one segments this forces the 120° law. More generally, the weighted tangent vectors must close up as a polygon.

2.4. Locally Bounded First Variation. Stationarity is the homogeneous equation $\delta V = 0$. A more flexible class is obtained by asking only that the first variation be representable by a Radon measure.

Definition 2.21. Let V be an n -varifold in an open set $U \subset \mathbb{R}^N$. We say that V has locally bounded first variation if for every $W \subset\subset U$ there is a constant $C(W) < \infty$ such that

$$|\delta V(X)| \leq C(W) \|X\|_{C^0(U)}$$

for every $X \in C_c^1(U; \mathbb{R}^N)$ with $\operatorname{supp} X \subset W$. In this case δV extends uniquely to an $\mathbb{R}^N$ -valued Radon measure on U .

The Radon–Nikodym theorem decomposes this vector measure relative to $\|V\|$. Write

$$\delta V = (\delta V)_{\text{ac}} + (\delta V)_s, \quad (\delta V)_{\text{ac}} \ll \|V\|, \quad (\delta V)_s \perp \|V\|.$$

Define

$$H_V = -\frac{d(\delta V)_{\text{ac}}}{d\|V\|}, \quad \sigma_V = |(\delta V)_s|, \quad \eta_V = \frac{d(\delta V)_s}{d\sigma_V}.$$

Then $H_V \in L^1_{\text{loc}}(\|V\|; \mathbb{R}^N)$ is the generalized mean curvature, σ_V is the generalized boundary measure, and $|\eta_V| = 1$ for σ_V -a.e. point; η_V is the generalized outward conormal. With these conventions,

$$\delta V(X) = - \int_U H_V \cdot X \, d\|V\| + \int_U X \cdot \eta_V \, d\sigma_V, \quad X \in C_c^1(U; \mathbb{R}^N).$$

This is the varifold version of the divergence theorem. The word “boundary” should be understood in this measure-theoretic sense: σ_V records the part of the first variation singular with respect to the mass measure, and need not coincide with a topological boundary.

Example 2.22 (The smooth case revisited). Let $M^n \subset U \subset \mathbb{R}^{n+m}$ be smooth with smooth boundary and set $V = |M|$. From the usual divergence theorem,

$$\delta|M|(X) = \int_{\partial M} X \cdot \eta \, d\mathcal{H}^{n-1} - \int_M H_M \cdot X \, d\mathcal{H}^n.$$

Comparing this with the Radon–Nikodym decomposition gives

$$H_V = H_M, \quad \sigma_V = \mathcal{H}^{n-1} \llcorner \partial M, \quad \eta_V = \eta,$$

where η is the outward unit conormal to ∂M in M . In particular, if $\partial M = \emptyset$ in U , then $\sigma_V = 0$ and the entire first variation is absolutely continuous with respect to $\|V\|$.

Example 2.23 (Two rays meeting at a corner). Let $R_1 = \{t\tau_1 : t \geq 0\}$ and $R_2 = \{t\tau_2 : t \geq 0\}$, where $\tau_1, \tau_2 \in \mathbb{R}^2$ are unit vectors, and set $V = |R_1| + |R_2|$. The interiors of the rays are straight, so there is no absolutely continuous curvature:

$$H_V = 0.$$

As in the Y -junction computation,

$$\delta V(X) = -X(0) \cdot (\tau_1 + \tau_2).$$

Hence, if $\tau_1 + \tau_2 \neq 0$,

$$\sigma_V = |\tau_1 + \tau_2| \delta_0, \quad \eta_V(0) = -\frac{\tau_1 + \tau_2}{|\tau_1 + \tau_2|}.$$

If $\tau_1 + \tau_2 = 0$, the two rays form a straight line with multiplicity one, and then $\sigma_V = 0$ and V is stationary.

Example 2.24 (Regular polygons converging to the circle). Let $N \geq 3$, let $p_k = \exp\left(\frac{2\pi i k}{N}\right) \in \mathbb{S}^1 \subset \mathbb{R}^2$, $k = 0, \dots, N-1$, and let M_N be the boundary of the regular N -gon with vertices p_k . Put $V_N = |M_N|$. On each open side $H_{V_N} = 0$, so the first variation is concentrated at the corners. If

$$\tau_k^+ = \frac{p_{k+1} - p_k}{|p_{k+1} - p_k|}, \quad \tau_k^- = \frac{p_{k-1} - p_k}{|p_{k-1} - p_k|},$$

then the vertex contribution is

$$-X(p_k) \cdot (\tau_k^+ + \tau_k^-).$$

For the regular polygon,

$$\tau_k^+ + \tau_k^- = -2 \sin\left(\frac{\pi}{N}\right) p_k.$$

Therefore

$$\delta V_N(X) = \sum_{k=0}^{N-1} 2 \sin\left(\frac{\pi}{N}\right) X(p_k) \cdot p_k,$$

and the generalized data are

$$H_{V_N} = 0, \quad \sigma_{V_N} = \sum_{k=0}^{N-1} 2 \sin\left(\frac{\pi}{N}\right) \delta_{p_k}, \quad \eta_{V_N}(p_k) = p_k.$$

As $N \rightarrow \infty$, the varifolds V_N converge to $V = |\mathbb{S}^1|$. Also

$$\eta_{V_N} \sigma_{V_N} \xrightarrow{*} x \mathcal{H}^1 \llcorner \mathbb{S}^1.$$

For the limiting circle,

$$\delta V(X) = \int_{\mathbb{S}^1} X \cdot x \, d\mathcal{H}^1.$$

Thus the limit has no generalized boundary:

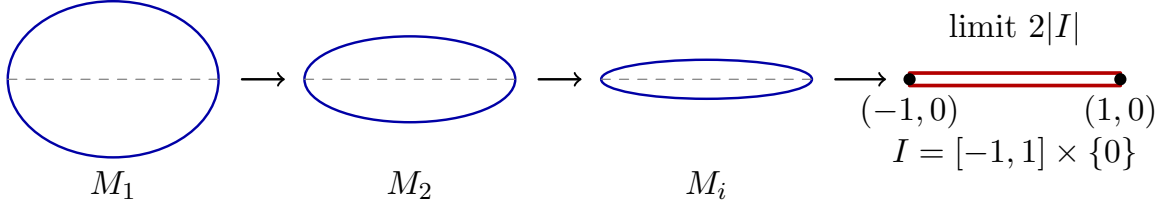
$$\sigma_V = 0, \quad H_V(x) = -x \quad \text{on } \mathbb{S}^1.$$

In this example the corner measures of the polygons converge to the ordinary curvature term of the smooth circle.

Example 2.25 (Flattening ellipses). Let $r_i \downarrow 0$ and let

$$M_i = \{(\cos t, r_i \sin t) : 0 \leq t < 2\pi\}, \quad V_i = |M_i|.$$

The width is fixed equal to 2, while the height is $2r_i$.



Each M_i is a smooth closed curve. Hence

$$\sigma_{V_i} = 0, \quad H_{V_i} = H_{M_i}.$$

With the parametrization above, if $s_i(t)^2 = \sin^2 t + r_i^2 \cos^2 t$, then the curvature vector is

$$H_{V_i}(t) = \frac{(-r_i^2 \cos t, -r_i \sin t)}{s_i(t)^4}.$$

Although these curvature vectors become large near the two tips, the varifolds have a simple limit:

$$V_i \rightarrow V = 2|I|, \quad I = [-1, 1] \times \{0\}.$$

The limiting varifold is a multiplicity-two segment. Its interior is straight, so

$$H_V = 0,$$

but now the first variation is supported at the endpoints:

$$\delta V(X) = 2X(1, 0) \cdot e_1 - 2X(-1, 0) \cdot e_1.$$

Thus

$$\sigma_V = 2\delta_{(1,0)} + 2\delta_{(-1,0)}, \quad \eta_V(1, 0) = e_1, \quad \eta_V(-1, 0) = -e_1.$$

Finally, the first variations converge weakly as vector-valued Radon measures, written $\delta V_i \xrightarrow{*} \delta V$. Indeed, for every $X \in C_c^1(\mathbb{R}^2; \mathbb{R}^2)$,

$$\delta V_i(X) = \int_{G_1(\mathbb{R}^2)} \operatorname{div}_S X(x) dV_i(x, S) \rightarrow \int_{G_1(\mathbb{R}^2)} \operatorname{div}_S X(x) dV(x, S) = \delta V(X),$$

because $V_i \rightarrow V$ and $(x, S) \mapsto \operatorname{div}_S X(x)$ is continuous and compactly supported. Thus a sequence with no generalized boundary can converge to a varifold whose first variation is purely boundary measure.

Theorem 2.26 (Allard compactness theorem [1, Section 6], [12, Theorem 42.7]). *Let $U \subset \mathbb{R}^N$ be open and let V_i be integral n -varifolds in U . Suppose that for every $W \subset\subset U$,*

$$\sup_i \|V_i\|(W) < \infty, \quad \sup_i |\delta V_i|(W) < \infty,$$

where $|\delta V_i|$ denotes the total variation measure of the vector-valued first variation. Then, after passing to a subsequence, there is an integral n -varifold V in U such that

$$V_i \rightarrow V \quad \text{as varifolds.}$$

Moreover,

$$\|V_i\| \xrightarrow{*} \|V\|$$

as Radon measures on U . Thus the class of integral varifolds with locally uniformly bounded mass and locally uniformly bounded first variation is compact, and integrality is preserved in the limit.

Remark 2.27 (Why the sawtooth example does not contradict Allard compactness). For the sawtooth varifolds $|M_i|$ in Example 2.8, the mass condition is satisfied:

$$\| |M_i| \|(\mathbb{R}^2) = 2\sqrt{a^2 + b^2}.$$

However, the first variations are not locally uniformly bounded. Let

$$\tau_+ = \frac{(b, a)}{\sqrt{a^2 + b^2}}, \quad \tau_- = \frac{(b, -a)}{\sqrt{a^2 + b^2}}.$$

At each interior corner the first variation has an atomic contribution whose size is

$$|\tau_+ - \tau_-| = \frac{2a}{\sqrt{a^2 + b^2}}.$$

There are $2^i - 1$ interior corners, so on any open set containing the whole sawtooth,

$$|\delta(|M_i|)|(\mathbb{R}^2) \geq (2^i - 1) \frac{2a}{\sqrt{a^2 + b^2}} \rightarrow \infty.$$

Hence the second hypothesis of Allard's compactness theorem fails. This must happen, because the varifold limit computed earlier carries two different tangent directions S_+ and S_- over the same line segment I ; for $a > 0$ it is not an integral varifold.

2.5. Some Drawbacks of Varifolds. Varifolds are extremely well adapted to variational problems: they have a compact weak topology, multiplicities are positive, the first variation is stable under weak convergence in many useful situations, and stationarity makes sense without parametrizations. Still, two limitations should be kept in mind.

First, boundary is not built into the object. One can define generalized boundary through the singular part of δV , as above, or restrict the class of variations near a prescribed set Γ , but this is not the same as having an algebraic boundary operator. The generalized boundary records force imbalance in the first variation; it does not by itself remember that a surface spans a fixed oriented curve.

This is why varifolds are excellent for interior regularity, while Plateau problems with a prescribed boundary often require extra structure.

Second, weak convergence of varifolds is a measure-theoretic notion, and this topology can be geometrically coarse. Two surfaces may be close as Radon measures while their shapes, topology, or Hausdorff behavior are very different. For instance, the earlier picture of a disk growing many long thin tubes should not be interpreted as a harmless smooth perturbation. If the tubes carry very small n -measure, they can disappear in the varifold limit even though the topology has changed dramatically; if they carry non-negligible area, they may destroy compactness or leave extra mass.

Thus varifolds see the distribution of area and tangent planes, not a full geometric shape in the classical sense.

3. CURRENTS

3.1. Basic Definitions. Currents add orientation to the rectifiable picture. This gives a natural boundary operator and allows cancellation, which is sometimes exactly what is needed in Plateau-type problems.

Definition 3.1. Let $U \subset \mathbb{R}^N$ be open. An integer multiplicity rectifiable n -current is written

$$T = \underline{\tau}(M, \theta, \vec{\tau}),$$

where $M \subset U$ is countably n -rectifiable, $\theta \in L^1_{\text{loc}}(\mathcal{H}^n \llcorner M; \mathbb{N})$ is the integer multiplicity, and $\vec{\tau}(x)$ is a measurable orientation of the approximate tangent plane. More explicitly, for $\mathcal{H}^n$ -a.e. $x \in M$, if $(e_1, \dots, e_n)$ is an oriented orthonormal basis of $T_x M$, then

$$\vec{\tau}(x) = e_1 \wedge \dots \wedge e_n$$

is a unit simple n -vector spanning $T_x M$. Thus T is the oriented analogue of the rectifiable varifold $\underline{\nu}(M, \theta)$. This definition does not yet impose any rectifiability condition on the boundary; the stricter notion of an integral current will be defined after we examine some examples of ∂T .

Example 3.2 (Smooth oriented manifolds). Let $M^n \subset U$ be an oriented C^1 submanifold, possibly with boundary. We write

$$[M] = \underline{\tau}(M, 1, \vec{\tau}_M).$$

Here $\vec{\tau}_M$ is determined by the given orientation. The notation $[M]$ should be read as the current obtained by integrating forms over M .

Let $\mathcal{D}^n(U)$ denote the space of compactly supported smooth n -forms in U . If $T = \underline{\tau}(M, \theta, \vec{\tau})$ and $\omega \in \mathcal{D}^n(U)$, then

$$T(\omega) = \int_M \theta(x) \langle \omega(x), \vec{\tau}(x) \rangle d\mathcal{H}^n(x) = \int_M \theta(x) \omega_x(\vec{\tau}(x)) d\mathcal{H}^n(x).$$

In particular, if $\vec{\tau} = e_1 \wedge \dots \wedge e_n$, then

$$\omega_x(\vec{\tau}) = \omega_x(e_1, \dots, e_n).$$

A general n -current is a continuous linear functional on $\mathcal{D}^n(U)$; rectifiable currents are the geometric examples among all such functionals.

Definition 3.3. The boundary of an n -current T is the $(n-1)$ -current

$$(\partial T)(\eta) = T(d\eta), \quad \eta \in \mathcal{D}^{n-1}(U).$$

For a smooth oriented manifold this is Stokes' theorem:

$$\partial[M] = [\partial M]$$

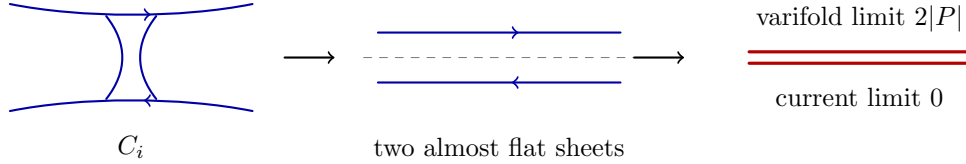
with the induced boundary orientation. A sequence T_i converges weakly to T as currents, written $T_i \rightharpoonup T$, if

$$T_i(\omega) \rightarrow T(\omega) \quad \text{for every } \omega \in \mathcal{D}^n(U).$$

Example 3.4 (Catenoid blow-down and cancellation). Let $C \subset \mathbb{R}^3$ be the standard oriented catenoid and let $C_i = r_i C$ with $r_i \downarrow 0$. Set $T_i = \llbracket C_i \rrbracket$. As varifolds, one sees two sheets collapsing onto the same plane $P = \{x_3 = 0\}$, hence $|T_i| \rightarrow 2|P|$ locally. As currents, however, the two sheets have opposite orientations. For every compactly supported 2-form ω ,

$$T_i(\omega) \rightarrow \int_P \omega(e_1 \wedge e_2) d\mathcal{H}^2 - \int_P \omega(e_1 \wedge e_2) d\mathcal{H}^2 = 0.$$

Thus $T_i \rightarrow 0$ as currents, while the associated varifolds converge to the multiplicity-two plane. The difference is cancellation.



Definition 3.5. If $T = \underline{\tau}(M, \theta, \vec{\tau})$, then its associated varifold is

$$|T| = \underline{v}(M, \theta),$$

obtained by forgetting the orientation. The mass measure of T is

$$\|T\| = \theta \mathcal{H}^n \llcorner M.$$

We define the support of T by the support of this mass measure:

$$\text{supp } T := \text{supp } \|T\| = \{x \in U : \|T\|(B_r(x)) > 0 \text{ for every } r > 0\}.$$

Equivalently, one can recover $\|T\|$ from the dual action of T . If $W \subset U$ is open, then

$$\|T\|(W) = \sup \{|T(\omega)| : \omega \in \mathcal{D}^n(U), \text{supp } \omega \subset W, |\omega(x)|_* \leq 1\},$$

where $|\omega(x)|_*$ is the comass norm, namely the supremum of $|\omega_x(\xi)|$ over all unit simple n -vectors ξ . The inequality “ $\leq$ ” follows from $|\omega_x(\vec{\tau}(x))| \leq 1$; equality follows by choosing test forms whose comass is at most one and which locally agree with the covector dual to $\vec{\tau}$ on M , then using a partition of unity.

Lemma 3.6 (Fatou lemma for the mass of currents). *If $T_i \rightarrow T$ weakly as n -currents in U , then for every open $W \subset U$,*

$$\|T\|(W) \leq \liminf_{i \rightarrow \infty} \|T_i\|(W).$$

In particular, $\mathbf{M}_W(T) \leq \liminf_i \mathbf{M}_W(T_i)$, where $\mathbf{M}_W(T) = \|T\|(W)$.

Proof. Fix $\omega \in \mathcal{D}^n(U)$ with $\text{supp } \omega \subset W$ and $|\omega|_* \leq 1$. By weak convergence,

$$|T(\omega)| = \lim_{i \rightarrow \infty} |T_i(\omega)| \leq \liminf_{i \rightarrow \infty} \|T_i\|(W).$$

Taking the supremum over all such ω gives the claim. $\square$

Example 3.7 (The sawtooth example as currents). Return to the sawtooth curves M_i from Example 2.8, and orient each segment from left to right. Let

$$T_i = \llbracket M_i \rrbracket, \quad I = [0, 2b] \times \{0\},$$

where I is oriented by e_1 . If $\omega = P dx + Q dy \in \mathcal{D}^1(\mathbb{R}^2)$, write the sawtooth as the graph $y = u_i(x)$. On alternating intervals the slope is $\pm \frac{a}{b}$, so

$$T_i(\omega) = \sum_{j=0}^{2^i-1} \int_{x_j}^{x_{j+1}} \left[P(x, u_i(x)) + (-1)^j \frac{a}{b} Q(x, u_i(x)) \right] dx, \quad x_j = \frac{bj}{2^{i-1}}.$$

Since $\|u_i\|_{C^0} \rightarrow 0$, the P -part converges to $\int_0^{2b} P(x, 0) dx$. The Q -part cancels between neighboring up and down segments; more precisely, pairing consecutive intervals gives an error bounded by $C\|DQ\|_{C^0}2^{-i}$, hence it tends to zero. Therefore

$$T_i \rightharpoonup \llbracket I \rrbracket, \quad \llbracket I \rrbracket(\omega) = \int_0^{2b} P(x, 0) dx.$$

The boundary is stable:

$$\partial T_i = \delta_B - \delta_O, \quad \partial \llbracket I \rrbracket = \delta_B - \delta_O,$$

because all interior vertices cancel algebraically. The masses are

$$\|T_i\| = \mathcal{H}^1 \llcorner M_i, \quad \mathbf{M}(T_i) = 2\sqrt{a^2 + b^2},$$

while

$$\|\llbracket I \rrbracket\| = \mathcal{H}^1 \llcorner I, \quad \mathbf{M}(\llbracket I \rrbracket) = 2b.$$

Thus the Fatou inequality is strict when $a > 0$:

$$2b < 2\sqrt{a^2 + b^2}.$$

Notice also that $|T_i| = |M_i|$ has the varifold limit computed earlier, whereas $|\lim T_i| = \|\llbracket I \rrbracket\| = |I|$. Passing to currents first allows cancellation, while passing to varifolds first retains the oscillating tangent directions and the excess mass.

This example is the promised contrast with varifolds. The limiting varifold in Example 2.8 is not integral when $a > 0$, because it keeps two distinct tangent directions over the same line segment. The current limit, however, is the perfectly integer multiplicity rectifiable current $\llbracket I \rrbracket$. This is one of the main advantages of currents: orientation permits cancellation, and cancellation often restores compactness in Plateau-type problems.

Proposition 3.8 (Boundary is continuous under weak convergence). *If $T_i \rightharpoonup T$ as n -currents in U , then*

$$\partial T_i \rightharpoonup \partial T$$

as $(n-1)$ -currents.

Proof. For every $\eta \in \mathcal{D}^{n-1}(U)$,

$$(\partial T_i)(\eta) = T_i(d\eta) \rightarrow T(d\eta) = (\partial T)(\eta).$$

This is exactly the definition of weak convergence of currents. □

This is very useful, but there is still a subtle point. The definition of ∂T always gives a current, because it is a distributional operation. It does not automatically imply that ∂T is rectifiable, or that it has locally finite mass. Thus an integer multiplicity rectifiable current may have a boundary as a current while that boundary is not yet a geometric object of the same class.

Example 3.9 (Top-dimensional currents in $\mathbb{R}^n$). Let $U \subset \mathbb{R}^n$ be open and let T be an integer multiplicity rectifiable n -current in U . Since the only unoriented n -plane in $\mathbb{R}^n$ is $\mathbb{R}^n$ itself, the orientation is just a sign. Thus there is an integer-valued function

$$\hat{\theta} \in L^1_{\text{loc}}(U; \mathbb{Z})$$

such that, for every $f \in C_c^\infty(U)$,

$$T(f dx^1 \wedge \cdots \wedge dx^n) = \int_U f \hat{\theta} d\mathcal{L}^n.$$

Indeed, if $T = \underline{T}(M, \theta, \vec{\tau})$, then $\mathcal{H}^n \llcorner M$ is absolutely continuous with respect to $\mathcal{L}^n$ and $\vec{\tau} = \pm e_1 \wedge \cdots \wedge e_n$ for $\mathcal{H}^n$ -a.e. $x \in M$; after extending by zero off M , one has

$$\widehat{\theta}(x) = \begin{cases} \theta(x), & \vec{\tau}(x) = e_1 \wedge \cdots \wedge e_n, \\ -\theta(x), & \vec{\tau}(x) = -e_1 \wedge \cdots \wedge e_n, \\ 0, & x \notin M. \end{cases}$$

Hence every top-dimensional integer multiplicity rectifiable current in $\mathbb{R}^n$ is just integration against an integer-valued L^1_{loc} density.

Let $\text{vol} = dx^1 \wedge \cdots \wedge dx^n$. For a compactly supported smooth vector field X , define the $(n-1)$ -form

$$*X = \iota_X \text{vol}.$$

Then $d(*X) = (\text{div } X)\text{vol}$, and therefore

$$\partial T(*X) = T(d(*X)) = \int_U \widehat{\theta} \text{div } X \, d\mathcal{L}^n.$$

Thus ∂T is the distributional derivative of $\widehat{\theta}$ in disguise:

$$\partial T(*X) = -D\widehat{\theta}(X).$$

If $\widehat{\theta}$ is very rough, this distribution need not be represented by a Radon measure, and then ∂T is not a rectifiable current. For instance, take $T = \llbracket E \rrbracket$ with $\widehat{\theta} = \chi_E$, where $E \subset U$ is measurable but has infinite perimeter in every relevant compact set. Then ∂T is the distributional boundary of E , but not an integer multiplicity rectifiable boundary current with locally finite mass.

This computation also gives the closedness condition:

$$\partial T = 0 \iff \int_U \widehat{\theta} \text{div } X \, d\mathcal{L}^n = 0 \text{ for all } X \in C_c^\infty(U; \mathbb{R}^n) \iff D\widehat{\theta} = 0.$$

Equivalently, $\widehat{\theta}$ is constant on each connected component of U . Thus a top-dimensional cycle in a connected open subset of $\mathbb{R}^n$ is just an integer multiple of the whole region.

Theorem 3.10 (Boundary rectifiability theorem [8]). *Let T be an integer multiplicity rectifiable n -current in an open set $U \subset \mathbb{R}^{n+m}$. If the boundary has locally finite mass,*

$$\|\partial T\|(W) < \infty \quad \text{for every } W \subset\subset U,$$

then ∂T is an integer multiplicity rectifiable $(n-1)$ -current. In particular, under the locally finite boundary-mass hypothesis, T satisfies exactly the extra boundary regularity required in the definition of integral currents below.

Returning to the top-dimensional example, this theorem says exactly that the bad behavior is ruled out by a perimeter condition. The identity

$$\partial T(*X) = \int_U \widehat{\theta} \text{div } X \, d\mathcal{L}^n$$

shows that $\|\partial T\|$ is locally finite precisely when $\widehat{\theta} \in BV_{\text{loc}}(U; \mathbb{Z})$. In that case $D\widehat{\theta}$ is a vector-valued Radon measure, and the boundary current is represented by the jump hypersurfaces of $\widehat{\theta}$. For $\widehat{\theta} = \chi_E$, this is the familiar finite-perimeter condition:

$$\chi_E \in BV_{\text{loc}}(U) \iff E \text{ has locally finite perimeter,}$$

and $\partial \llbracket E \rrbracket$ is integration over the reduced boundary $\partial^* E$ with the measure-theoretic normal orientation. This is the current analogue of saying that the boundary has finite length or finite area.

3.2. Examples of Current Boundaries.

Example 3.11 (Smooth oriented submanifolds and Stokes' theorem). Let $M^n \subset U$ be an oriented C^1 submanifold with C^1 boundary. As mentioned above,

$$\partial \llbracket M \rrbracket = \llbracket \partial M \rrbracket.$$

Indeed, for every $\eta \in \mathcal{D}^{n-1}(U)$,

$$(\partial \llbracket M \rrbracket)(\eta) = \llbracket M \rrbracket(d\eta) = \int_M d\eta = \int_{\partial M} \eta = \llbracket \partial M \rrbracket(\eta),$$

which is exactly Stokes' theorem with the induced boundary orientation.

The one-dimensional special case is particularly concrete. Let $\gamma : [a, b] \rightarrow \mathbb{R}^2$ be an oriented regular C^1 curve with initial point $L = \gamma(a)$ and terminal point $N = \gamma(b)$, and set $T = \llbracket \gamma \rrbracket$. For $f \in C_c^\infty(\mathbb{R}^2) = \mathcal{D}^0(\mathbb{R}^2)$,

$$(\partial T)(f) = T(df) = \int_a^b df_{\gamma(t)}(\gamma'(t)) dt = f(\gamma(b)) - f(\gamma(a)).$$

Thus

$$\partial \llbracket \gamma \rrbracket = \delta_N - \delta_L,$$

the current version of “terminal point minus initial point.”

Example 3.12 (Two oriented segments meeting at a corner). Let $p, q_1, q_2 \in \mathbb{R}^2$, and let $E_i = [p, q_i]$ be oriented away from the common point p . Set

$$T = \llbracket E_1 \rrbracket + \llbracket E_2 \rrbracket.$$

If $f \in C_c^\infty(\mathbb{R}^2)$, the previous curve computation gives

$$(\partial T)(f) = (f(q_1) - f(p)) + (f(q_2) - f(p)).$$

Hence

$$\partial T = \delta_{q_1} + \delta_{q_2} - 2\delta_p.$$

The corner point is counted twice because, with this orientation choice, both segments start at p . If one segment were oriented into p and one out of p , the coefficients at p would cancel instead. Thus the current boundary is an algebraic incidence count, not a curvature or angle defect.

Example 3.13 (Three 120° rays: current boundary versus varifold boundary). Let $\tau_1, \tau_2, \tau_3 \in \mathbb{R}^2$ be unit vectors making mutual angles 120° , so that

$$\tau_1 + \tau_2 + \tau_3 = 0.$$

Let

$$R_i = \{p + t\tau_i : t \geq 0\}$$

and orient each ray away from the common point p . Define the locally finite current

$$T = \llbracket R_1 \rrbracket + \llbracket R_2 \rrbracket + \llbracket R_3 \rrbracket.$$

For $f \in C_c^\infty(\mathbb{R}^2)$,

$$(\partial \llbracket R_i \rrbracket)(f) = \int_0^\infty \frac{d}{dt} f(p + t\tau_i) dt = -f(p),$$

because f vanishes for large t . Therefore

$$(\partial T)(f) = -3f(p), \quad \partial T = -3\delta_p.$$

So the common point produces a current boundary of multiplicity -3 .

From the varifold viewpoint the same support behaves differently. Let

$$V = |R_1| + |R_2| + |R_3|.$$

For a compactly supported vector field X ,

$$\delta V(X) = - \sum_{i=1}^3 X(p) \cdot \tau_i = -X(p) \cdot (\tau_1 + \tau_2 + \tau_3) = 0.$$

Thus this is a stationary varifold and has no generalized boundary in the first-variation sense. The reason for the difference is that varifold stationarity tests vector balance of the tangent directions, while the current boundary tests oriented incidence. Three outgoing oriented rays do not cancel algebraically, even though their tangent vectors balance geometrically.

3.3. Integral Currents.

Definition 3.14. Let $U \subset \mathbb{R}^N$ be open. An integer multiplicity rectifiable n -current T in U is called an integral n -current in U if its boundary

$$\partial T$$

is an integer multiplicity rectifiable $(n-1)$ -current in U . Equivalently, T is integer multiplicity rectifiable, $\|T\|$ is locally finite, and $\|\partial T\|$ is locally finite; then the boundary rectifiability theorem implies that ∂T is again integer multiplicity rectifiable. We write this class as

$$\mathcal{I}_n(U).$$

Thus

$$\begin{aligned} T \in \mathcal{I}_n(U) &\iff T \text{ is an integer multiplicity rectifiable } n\text{-current,} \\ &\quad \partial T \text{ is an integer multiplicity rectifiable } (n-1)\text{-current.} \end{aligned}$$

The word “integral” is therefore strictly stronger than merely integer multiplicity rectifiable: it includes a geometric boundary condition. Smooth oriented manifolds with smooth boundary are integral currents by Stokes’ theorem, and the line segment current $\llbracket I \rrbracket$ above is integral because its boundary is the difference of two point masses. By contrast, the rough top-dimensional current with density $\hat{\theta} \notin BV_{\text{loc}}$ is integer multiplicity rectifiable but not integral, since its boundary is only a distributional current and not a rectifiable current with locally finite mass.

Theorem 3.15 (Federer–Fleming compactness theorem [8]). *Let $U \subset \mathbb{R}^N$ be open and let $T_i \in \mathcal{I}_n(U)$. Suppose that for every $W \subset\subset U$,*

$$\sup_i (\|T_i\|(W) + \|\partial T_i\|(W)) < \infty.$$

Then there are a subsequence, still denoted T_i , and an integral current $T \in \mathcal{I}_n(U)$ such that

$$T_i \rightharpoonup T$$

as currents. Consequently,

$$\partial T_i \rightharpoonup \partial T.$$

In words: locally uniform mass bounds for both the currents and their boundaries give compactness inside the class of integral currents.

3.4. Plateau Problem for Integral Currents. Let $\Gamma \in \mathcal{I}_{n-1}(\mathbb{R}^N)$ be a compactly supported integral cycle, meaning

$$\partial \Gamma = 0.$$

The current version of the Plateau problem is to minimize mass among all integral currents spanning Γ :

$$\begin{aligned} \mathcal{I}(\Gamma) &= \{T \in \mathcal{I}_n(\mathbb{R}^N) : \partial T = \Gamma, \mathbf{M}(T) < \infty\}, \\ \text{find } T_* \in \mathcal{I}(\Gamma) \text{ such that } \mathbf{M}(T_*) &= \inf_{T \in \mathcal{I}(\Gamma)} \mathbf{M}(T). \end{aligned}$$

First one needs to know that $\mathcal{I}(\Gamma)$ is nonempty. This is supplied by the cone construction. Choose a point $p \in \mathbb{R}^N$ and define

$$F : [0, 1] \times \mathbb{R}^N \rightarrow \mathbb{R}^N, \quad F(t, x) = p + t(x - p).$$

The cone over Γ with vertex p is

$$C_p\Gamma := F_{\#}([0, 1] \times \Gamma).$$

Products and Lipschitz pushforwards preserve integral currents, so $C_p\Gamma \in \mathcal{I}_n(\mathbb{R}^N)$. The boundary formula for products and pushforwards gives

$$\partial(C_p\Gamma) = F_{\#}([1] \times \Gamma - [0] \times \Gamma - [0, 1] \times \partial\Gamma) = \Gamma - p_{\#}\Gamma.$$

Since $\partial\Gamma = 0$, the last term involving $\partial\Gamma$ vanishes. If $n \geq 2$, the constant pushforward $p_{\#}\Gamma$ is zero for dimensional reasons; for $n = 1$ one assumes the usual zero total multiplicity condition on the 0-cycle Γ . Thus $\partial(C_p\Gamma) = \Gamma$, and $C_p\Gamma \in \mathcal{I}(\Gamma)$. Moreover, if $\text{supp } \Gamma \subset B_R(p)$, the standard cone estimate gives

$$\mathbf{M}(C_p\Gamma) \leq \frac{R}{n} \mathbf{M}(\Gamma),$$

so the cone has finite mass.

Now the direct method applies. Let

$$m = \inf_{T \in \mathcal{I}(\Gamma)} \mathbf{M}(T),$$

and choose a minimizing sequence $T_i \in \mathcal{I}(\Gamma)$ with $\mathbf{M}(T_i) \rightarrow m$. Then

$$\sup_i \mathbf{M}(T_i) < \infty, \quad \mathbf{M}(\partial T_i) = \mathbf{M}(\Gamma),$$

so the compactness theorem gives a subsequence and an integral current T_* such that

$$T_i \rightarrow T_*.$$

By continuity of the boundary under current convergence,

$$\partial T_i \rightarrow \partial T_*.$$

Since $\partial T_i = \Gamma$ for every i , the same sequence also converges to Γ ; hence $\partial T_* = \Gamma$, and $T_* \in \mathcal{I}(\Gamma)$. Finally, the Fatou lower semicontinuity lemma for mass gives

$$\mathbf{M}(T_*) \leq \liminf_{i \rightarrow \infty} \mathbf{M}(T_i) = m.$$

Since m is the infimum over the admissible class, $\mathbf{M}(T_*) = m$. Thus the Plateau problem has a mass-minimizing integral current solution.

Remark 3.16 (Uniqueness and the next question). The minimizer T_* found by the direct method is not forced to be unique. The argument only proves that the bottom of the mass functional is attained; it gives no rule selecting one minimizer if two different integral currents with the same boundary have the same mass. Geometrically, symmetries or calibrated competitors may make the mass functional have a flat bottom.

There are nevertheless important uniqueness results. Here “generic” means generic in the Baire category sense: the good boundary data form a residual set. In this sense the Plateau problem is generically uniquely solvable, and the unique minimizer has multiplicity one [4].

In a cylindrical or graphical situation one also often has a more elementary uniqueness mechanism. For scalar graphs with fixed boundary values, the area integrand

$$p \mapsto \sqrt{1 + |p|^2}$$

is strictly convex in p , so the graphical minimizer is unique. This should be read as uniqueness inside the graphical or cylindrical class, not as an unrestricted uniqueness theorem for all currents.

The next natural question is regularity: how smooth is the minimizer T_* ? This is the regularity theory of area-minimizing currents, and we postpone it for later.

3.5. Sets of Locally Finite Perimeter. Sets of locally finite perimeter, also called Caccioppoli sets, are the codimension-one, multiplicity-one special case of the current picture. Instead of allowing an arbitrary integer density $\hat{\theta}$, we take the density to be the characteristic function χ_E of a measurable set.

Definition 3.17. Let $U \subset \mathbb{R}^n$ be open and let $E \subset U$ be $\mathcal{L}^n$ -measurable. For an open set $W \subset U$, define

$$P(E; W) := \sup \left\{ \int_E \operatorname{div} X \, d\mathcal{L}^n : X \in C_c^1(W; \mathbb{R}^n), \|X\|_{C^0} \leq 1 \right\}.$$

We say that E has finite perimeter in W if $P(E; W) < \infty$, and that E has locally finite perimeter in U if $P(E; W) < \infty$ for every $W \subset\subset U$. If $U = \mathbb{R}^n$ and $P(E; \mathbb{R}^n) < \infty$, we simply say that E has finite perimeter.

This definition is exactly what one expects from smooth domains. If E has C^1 boundary and $\operatorname{supp} X \subset W$, the divergence theorem gives

$$\int_E \operatorname{div} X \, d\mathcal{L}^n = \int_{\partial E} \langle X, \nu_E \rangle \, d\mathcal{H}^{n-1},$$

where ν_E is the outward unit normal. Hence

$$\left| \int_E \operatorname{div} X \, d\mathcal{L}^n \right| \leq \mathcal{H}^{n-1}(\partial E \cap W) \|X\|_{C^0}.$$

Thus $P(E; W) \leq \mathcal{H}^{n-1}(\partial E \cap W)$ for smooth E , and in fact equality holds with this convention. The definition therefore measures the size of the boundary without requiring a classical boundary.

The same definition can be translated directly into currents. Give $\mathbb{R}^n$ its standard orientation and define the multiplicity-one top-dimensional current

$$[E](f \, dx^1 \wedge \cdots \wedge dx^n) := \int_E f \, d\mathcal{L}^n.$$

For a compactly supported vector field X , set

$$*X := \iota_X(dx^1 \wedge \cdots \wedge dx^n).$$

Then $*X$ is an $(n-1)$ -form, $d(*X) = (\operatorname{div} X) \, dx^1 \wedge \cdots \wedge dx^n$, and therefore

$$\partial[E](*X) = [E](d(*X)) = \int_E \operatorname{div} X \, d\mathcal{L}^n.$$

Since the Hodge-star identification $X \mapsto *X$ preserves the pointwise norm and comass in Euclidean space, taking suprema over $|X| \leq 1$ is the same as taking the mass norm of the boundary current. Consequently

$$P(E; W) = \|\partial[E]\|(W).$$

Thus E has locally finite perimeter in U if and only if the current boundary $\partial[E]$ has locally finite mass in U . In that case $[E]$ is an integral n -current locally in U , and $\partial[E]$ is the current carried by the measure-theoretic boundary of E .

Theorem 3.18 (De Giorgi structure theorem [12]). *Let $E \subset U \subset \mathbb{R}^n$ be a set of locally finite perimeter. Then there is a countably $(n-1)$ -rectifiable set $\partial^*E \subset U$, called the reduced boundary, and a Borel unit vector field*

$$\eta_E : \partial^*E \rightarrow \mathbb{S}^{n-1},$$

called the measure-theoretic outward conormal, such that for every $X \in C_c^1(U; \mathbb{R}^n)$,

$$\int_E \operatorname{div} X \, d\mathcal{L}^n = \int_{\partial^*E} \langle X, \eta_E \rangle \, d\mathcal{H}^{n-1}.$$

Equivalently,

$$-D\chi_E = \eta_E \mathcal{H}^{n-1} \llcorner \partial^*E, \quad |D\chi_E| = \mathcal{H}^{n-1} \llcorner \partial^*E.$$

Moreover, for $\mathcal{H}^{n-1}$ -a.e. $x \in \partial^* E$, the approximate tangent plane exists and is

$$T_x(\partial^* E) = \eta_E(x)^\perp,$$

and the blow-ups of E converge in L_{loc}^1 to the half-space with outward normal $\eta_E(x)$:

$$\chi_{\frac{E-x}{r}} \rightarrow \chi_{\{y \cdot \eta_E(x) < 0\}} \quad \text{in } L_{\text{loc}}^1(\mathbb{R}^n) \quad \text{as } r \downarrow 0.$$

Thus a finite-perimeter set has a genuine tangent hyperplane and a genuine normal at almost every boundary point, even though the topological boundary may be much larger.

With the boundary orientation determined by the convention that $\eta_E \wedge \vec{\tau}_E$ gives the standard orientation of $\mathbb{R}^n$, the current carried by the reduced boundary is

$$[\partial^* E](\omega) := \int_{\partial^* E} \omega(\vec{\tau}_E) d\mathcal{H}^{n-1}.$$

The preceding theorem says exactly

$$\partial[E] = [\partial^* E].$$

Indeed, testing both currents against $*X = \iota_X(dx^1 \wedge \cdots \wedge dx^n)$ gives the same number

$$\int_{\partial^* E} \langle X, \eta_E \rangle d\mathcal{H}^{n-1}.$$

Definition 3.19. For a Borel set $A \subset U$, define

$$P(E; A) := |D\chi_E|(A).$$

By De Giorgi's theorem this is the same as

$$P(E; A) = \mathcal{H}^{n-1}(\partial^* E \cap A) = \|\partial[E]\|(A).$$

When $U = \mathbb{R}^n$ and $A = \mathbb{R}^n$, we write

$$P(E) := P(E; \mathbb{R}^n)$$

and call this the perimeter of E .

Theorem 3.20 (Compactness for finite-perimeter sets). *Let $U \subset \mathbb{R}^n$ be open and let $E_i \subset U$ be sets of locally finite perimeter. Suppose that for every $W \subset \subset U$,*

$$\sup_i P(E_i; W) < \infty.$$

Then there are a subsequence, still denoted E_i , and a set $E \subset U$ of locally finite perimeter such that

$$\chi_{E_i} \rightarrow \chi_E \quad \text{in } L_{\text{loc}}^1(U).$$

Equivalently,

$$[E_i] \rightharpoonup [E] \quad \text{as } n\text{-currents in } U,$$

and

$$\partial[E_i] \rightharpoonup \partial[E].$$

Moreover, for every open $W \subset U$,

$$P(E; W) \leq \liminf_{i \rightarrow \infty} P(E_i; W).$$

Proof. Set $S_i = [E_i]$. On every $W \subset U$,

$$\|S_i\|(W) = \mathcal{L}^n(E_i \cap W) \leq \mathcal{L}^n(W), \quad \|\partial S_i\|(W) = P(E_i; W),$$

so the Federer–Fleming compactness theorem gives a subsequence and an integral n -current S such that

$$S_i \rightharpoonup S.$$

Because S is top-dimensional in $U \subset \mathbb{R}^n$, it has the form

$$S(f \, dx^1 \wedge \cdots \wedge dx^n) = \int_U \theta f \, d\mathcal{L}^n$$

for some integer-valued $\theta \in L^1_{\text{loc}}(U)$. Since $0 \leq \chi_{E_i} \leq 1$, testing against nonnegative functions gives $0 \leq \theta \leq 1$ a.e.; hence $\theta = \chi_E$ for a measurable set E . Thus $S = \llbracket E \rrbracket$.

The statement $\chi_{E_i} \rightarrow \chi_E$ in L^1_{loc} is the set version of the same compactness. More explicitly, the bounds on $P(E_i; W)$ say that the functions χ_{E_i} are uniformly bounded in $BV(W)$. The compact embedding

$$BV(W) \hookrightarrow L^1(W') \quad \text{for } W' \subset\subset W$$

then gives an L^1 -convergent subsequence. Its distributional limit must be the same χ_E obtained from current convergence.

Finally, the boundary convergence follows from continuity of ∂ under current convergence, and lower semicontinuity is just the Fatou lemma for current mass:

$$P(E; W) = \|\partial \llbracket E \rrbracket\|(W) \leq \liminf_{i \rightarrow \infty} \|\partial \llbracket E_i \rrbracket\|(W) = \liminf_{i \rightarrow \infty} P(E_i; W).$$

□

3.6. Codimension-One Plateau Problem. The finite-perimeter formulation is the codimension-one version of Plateau's problem. Here the unknown hypersurface is not written first as an abstract current T ; instead it is written as the reduced boundary $\partial^* E$ of a region E . The boundary condition is therefore encoded by an exterior phase.

Let $U \subset \mathbb{R}^n$ be a bounded open set with Lipschitz boundary, and let $E_0 \subset \mathbb{R}^n$ be a set of locally finite perimeter. Define the admissible class

$$\mathcal{A}(E_0, U) := \{E \subset \mathbb{R}^n : E \text{ has locally finite perimeter and } E \setminus U = E_0 \setminus U \text{ up to } \mathcal{L}^n\text{-null sets}\}.$$

The relative Plateau problem is

$$\min\{P(E; U) : E \in \mathcal{A}(E_0, U)\}.$$

This differs slightly from the current formulation $\partial T = \Gamma$. In the current formulation the boundary Γ is an explicit $(n-2)$ -current. In the finite-perimeter formulation, the trace of χ_{E_0} on ∂U is prescribed. If ∂U is smooth and this trace jumps across a smooth $(n-2)$ -submanifold $\Gamma \subset \partial U$, then Γ is the boundary curve that the hypersurface $\partial^* E \cap U$ spans. Thus Γ appears as the interface on ∂U between the prescribed outside phase E_0 and its complement.

Existence follows from the compactness theorem above. The class is nonempty because $E_0 \in \mathcal{A}(E_0, U)$. Let E_i be a minimizing sequence. Since $E_i = E_0$ outside U and U has Lipschitz boundary, the perimeter of E_i is locally uniformly controlled near ∂U by the relative perimeter in U , the fixed perimeter of E_0 outside U , and the area of ∂U . Thus the compactness theorem gives, after passing to a subsequence,

$$\chi_{E_i} \rightarrow \chi_E \quad \text{in } L^1_{\text{loc}}(\mathbb{R}^n).$$

The exterior condition passes to the limit, so $E \in \mathcal{A}(E_0, U)$, and lower semicontinuity gives

$$P(E; U) \leq \liminf_{i \rightarrow \infty} P(E_i; U).$$

Hence E minimizes perimeter in U . The resulting current

$$(\partial \llbracket E \rrbracket) \llcorner U = \llbracket \partial^* E \cap U \rrbracket$$

is the current-theoretic hypersurface solving this codimension-one Plateau problem.

Proposition 3.21 (Codimension-one cycles are built from Caccioppoli boundaries). *Let $T \in \mathcal{I}_{n-1}(\mathbb{R}^n)$ be a compactly supported integral cycle, so $\partial T = 0$. Then there exist finite-perimeter sets $E_k, F_k \subset \mathbb{R}^n$ such that*

$$\sum_{k=1}^{\infty} (P(E_k) + P(F_k)) < \infty$$

and

$$T = \sum_{k=1}^{\infty} \partial \llbracket E_k \rrbracket - \sum_{k=1}^{\infty} \partial \llbracket F_k \rrbracket$$

as currents. In particular, the finite partial sums are boundaries of finite sums of Caccioppoli sets and converge to T as currents; with the natural truncations below, the tails also go to zero in mass.

Idea of the proof. Since $\mathbb{R}^n$ has no homology in dimension $n - 1$, the isoperimetric theorem for integral currents gives an integral n -current S with

$$\partial S = T.$$

Top-dimensional integral currents in $\mathbb{R}^n$ are exactly integer-valued BV densities:

$$S(f \, dx^1 \wedge \cdots \wedge dx^n) = \int_{\mathbb{R}^n} \theta f \, d\mathcal{L}^n, \quad \theta \in BV_c(\mathbb{R}^n; \mathbb{Z}).$$

Use the layer-cake decomposition of the integer-valued function θ :

$$E_k = \{\theta \geq k\}, \quad F_k = \{\theta \leq -k\}, \quad k = 1, 2, \dots$$

Then

$$\theta = \sum_{k=1}^{\infty} \chi_{E_k} - \sum_{k=1}^{\infty} \chi_{F_k} \quad \text{in } L^1,$$

and the BV coarea formula gives

$$\sum_{k=1}^{\infty} (P(E_k) + P(F_k)) = |D\theta|(\mathbb{R}^n) < \infty.$$

Therefore

$$S = \sum_{k=1}^{\infty} \llbracket E_k \rrbracket - \sum_{k=1}^{\infty} \llbracket F_k \rrbracket,$$

and applying ∂ gives the desired decomposition of T . Thus in codimension one, Caccioppoli boundaries are not merely examples; they are the basic building blocks of integral cycles. $\square$

4. REGULARITY OF AREA-MINIMIZING CURRENTS

4.1. Interior Stationarity of a Minimizer. We now return to the Plateau minimizer constructed above and ask about its regularity. To keep the setting clear, let

$$\Gamma^{n-1} \subset \mathbb{R}^{n+m}$$

be a compact oriented smooth submanifold without boundary, and let $\llbracket \Gamma \rrbracket$ be the associated integral $(n - 1)$ -cycle. The integral-current Plateau problem asks for

$$T \in \mathcal{I}_n(\mathbb{R}^{n+m}), \quad \partial T = \llbracket \Gamma \rrbracket,$$

such that

$$\mathbf{M}(T) = \|T\|(\mathbb{R}^{n+m}) = \inf \{ \mathbf{M}(S) : S \in \mathcal{I}_n(\mathbb{R}^{n+m}), \partial S = \llbracket \Gamma \rrbracket \}.$$

Such a T is called an area-minimizing integral current with boundary Γ . The regularity question is: after forgetting the weak formalism used to construct T , how much of T is actually a smooth minimal submanifold?

Before discussing smoothness, one should first record the Euler–Lagrange equation. Since the boundary is fixed, only variations supported away from Γ are allowed. The conclusion is that the varifold associated to T is stationary in the complement of the boundary.

Definition 4.1. Let $f : U \rightarrow U'$ be a proper Lipschitz map on the support of the current under consideration. If T is an n -current in U , its pushforward is the n -current $f_{\#}T$ in U' defined by

$$(f_{\#}T)(\omega) := T(f^{\#}\omega), \quad \omega \in \mathcal{D}^n(U').$$

Here $f^{\#}\omega$ is the pullback form. This notation is compatible with the boundary operator:

$$\partial(f_{\#}T) = f_{\#}(\partial T),$$

because exterior differentiation commutes with pullback. Indeed,

$$\partial(f_{\#}T)(\eta) = f_{\#}T(d\eta) = T(f^{\#}d\eta) = T(d(f^{\#}\eta)) = \partial T(f^{\#}\eta) = f_{\#}(\partial T)(\eta).$$

When f is a C^1 diffeomorphism and $T = \underline{\tau}(M, \theta, \vec{\tau})$, this is the current obtained by transporting the oriented rectifiable set:

$$f_{\#}T = \underline{\tau}(f(M), \theta \circ f^{-1}, f_{\#}\vec{\tau}),$$

and its associated varifold satisfies

$$|f_{\#}T| = f_{\#}|T|.$$

Proposition 4.2 (Interior stationarity). *Let $T \in \mathcal{I}_n(\mathbb{R}^{n+m})$ be area-minimizing with*

$$\partial T = \llbracket \Gamma \rrbracket.$$

Then the associated integral varifold

$$V := |T|$$

is stationary in $\mathbb{R}^{n+m} \setminus \Gamma$. Equivalently, for every vector field

$$X \in C_c^1(\mathbb{R}^{n+m} \setminus \Gamma; \mathbb{R}^{n+m}),$$

one has

$$\delta V(X) = 0.$$

Proof. Let

$$U := \mathbb{R}^{n+m} \setminus \Gamma$$

and fix $X \in C_c^1(U; \mathbb{R}^{n+m})$. Since $\text{supp } X$ is compact in U , there is an open neighborhood N of Γ on which $X = 0$. Let Φ_t be the flow generated by X :

$$\frac{d}{dt}\Phi_t(x) = X(\Phi_t(x)), \quad \Phi_0 = \text{id}.$$

For $|t|$ small, Φ_t is a C^1 diffeomorphism of $\mathbb{R}^{n+m}$ and $\Phi_t = \text{id}$ on N . Therefore the current pushforward $(\Phi_t)_{\#}T$ has the same boundary as T :

$$\partial((\Phi_t)_{\#}T) = (\Phi_t)_{\#}(\partial T) = (\Phi_t)_{\#}\llbracket \Gamma \rrbracket = \llbracket \Gamma \rrbracket.$$

Thus $(\Phi_t)_{\#}T$ is an admissible competitor in the same Plateau class.

By the minimizing property of T ,

$$\mathbf{M}(T) \leq \mathbf{M}((\Phi_t)_{\#}T)$$

for all small t . Because Φ_t is a diffeomorphism, pushing forward the current does not create cancellation; it only changes tangent Jacobians. Hence

$$\mathbf{M}((\Phi_t)_{\#}T) = \|(\Phi_t)_{\#}T\|(\mathbb{R}^{n+m}).$$

Set

$$F(t) := \|(\Phi_t)_{\#}V\|(\mathbb{R}^{n+m}).$$

The previous inequality says that $F(t) \geq F(0)$ for all small t , so $t = 0$ is a local minimum of F . Therefore

$$0 = F'(0).$$

By the definition of the first variation of a varifold,

$$F'(0) = \delta V(X) = \int_{G_n(\mathbb{R}^{n+m})} \operatorname{div}_S X(x) dV(x, S).$$

Thus $\delta V(X) = 0$ for every compactly supported vector field in $\mathbb{R}^{n+m} \setminus \Gamma$, which is precisely stationarity of $|T|$ away from the boundary. $\square$

In smooth language, this proposition says exactly what one expects. If the minimizer is represented near an interior point by a smooth embedded submanifold M , then $|T|$ is the multiplicity-weighted varifold $q|M|$, and stationarity gives

$$0 = \delta(q|M|)(X) = -q \int_M \langle H_M, X \rangle d\mathcal{H}^n$$

for all compactly supported X in that neighborhood. Hence $H_M = 0$. The hard part of regularity theory is the converse direction: starting only from an area-minimizing current, determine where it is forced to be a smooth minimal submanifold and how large the singular set can be.

4.2. Regularity Theorems and Model Singularities. We now state the interior regularity goal in the form that will guide the rest of the discussion. Let $T \in \mathcal{I}_n(\mathbb{R}^{n+m})$ be area-minimizing with smooth boundary $\partial T = \llbracket \Gamma \rrbracket$. A point $x \in \operatorname{supp} T \setminus \Gamma$ is called a regular point if, in some ball $B_\rho(x)$,

$$T \llcorner B_\rho(x) = q \llbracket M \rrbracket$$

for a positive integer q and a smooth embedded connected minimal n -submanifold $M \subset B_\rho(x)$. The regular set and singular set are

$$\operatorname{reg} T := \{x : x \text{ is regular}\}, \quad \operatorname{sing} T := (\operatorname{supp} T \setminus \Gamma) \setminus \operatorname{reg} T.$$

Theorem 4.3 (Interior regularity of area minimizers). *Let $T \in \mathcal{I}_n(\mathbb{R}^{n+m})$ be area-minimizing with smooth boundary $\llbracket \Gamma \rrbracket$. Then $\operatorname{reg} T$ is relatively open in $\operatorname{supp} T \setminus \Gamma$, and on $\operatorname{reg} T$ the current is a smooth embedded minimal submanifold with locally constant integer multiplicity.*

There are two basic size estimates for the singular set.

- (1) *If $m = 1$, so T is a codimension-one area-minimizing hypersurface, then*

$$\dim_{\mathcal{H}} \operatorname{sing} T \leq n - 7.$$

In particular, there are no interior singularities for $n \leq 6$.

- (2) *In arbitrary codimension,*

$$\dim_{\mathcal{H}} \operatorname{sing} T \leq n - 2.$$

The codimension-one estimate is due to the De Giorgi–Federer–Fleming–Simons theory and is sharp because of the Simons cone. The arbitrary-codimension estimate is Almgren’s big regularity theorem [7, 2, 12].

Example 4.4 (The Simons cone). The Simons cone is the hypersurface

$$C = \{(x, y) \in \mathbb{R}^4 \times \mathbb{R}^4 : |x| = |y|\} \subset \mathbb{R}^8.$$

It has dimension 7 and codimension 1. It is smooth away from the origin and singular at the origin. Simons showed the importance of such cones in the study of stable minimal hypersurfaces, and Bombieri–De Giorgi–Giusti proved that this cone is area-minimizing [13, 3]. Thus the codimension-one theorem cannot be improved beyond the threshold $n = 7$: singular area-minimizing hypersurfaces really do occur in dimension 7.

Example 4.5 (A complex cusp). Consider the complex curve

$$\mathcal{C} = \{(z, w) \in \mathbb{C}^2 : w^3 = z^2\}.$$

It can be parametrized by

$$z = t^3, \quad w = t^2, \quad t \in \mathbb{C}.$$

Away from $t = 0$ it is a smooth complex curve, while at the origin it has a cusp or branch singularity. Formally one may write

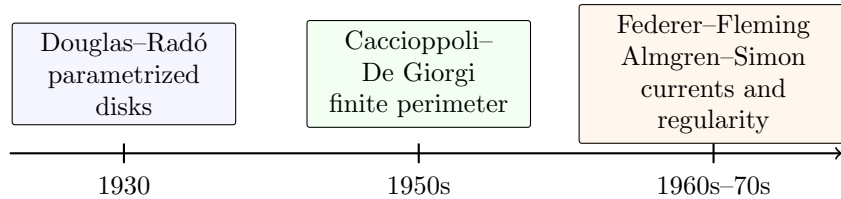
$$z = \pm w^{\frac{3}{2}},$$

which shows the branching over the w -plane. Since complex analytic subvarieties are calibrated by powers of the Kähler form, $\mathcal{C}$ is area-minimizing as a real 2-current in $\mathbb{R}^4$. Here $n = 2$ and the singular set is a point, so its dimension is $0 = n - 2$. This illustrates why the higher-codimension estimate $\dim_{\mathcal{H}} \text{sing } T \leq n - 2$ is natural and sharp in spirit.

Remark 4.6 (A short historical map). The word “solution” to Plateau’s problem depends on the weak category one chooses.

- Douglas and Radó solved the classical disk problem around 1930: for a Jordan curve in $\mathbb{R}^3$, they found a conformal harmonic parametrized disk spanning the curve [11, 6]. This is the parametric solution.
- Caccioppoli and De Giorgi developed the theory of sets of finite perimeter in the 1950s. In codimension one, one minimizes perimeter among regions with prescribed exterior data.
- Federer and Fleming introduced normal and integral currents in 1960, giving a flexible oriented theory with compactness and lower semicontinuity [8]. The later regularity theory of Federer, Fleming, Almgren, Simons, and Simon explains how smooth such minimizers are away from a small singular set.

Thus different frameworks find different kinds of weak surfaces: parametrized disks, reduced boundaries of Caccioppoli sets, and integral currents. They agree with the same classical minimal surface when the solution is smooth, but they behave differently when branching, multiplicity, cancellation, or singularities appear.



4.3. Regularity of Stationary Varifolds. The first quantitative tool for regularity is the monotonicity formula. It uses only stationarity, not minimizing. Let $V = \underline{v}(M, \theta)$ be an integral stationary n -varifold in an open set $U \subset \mathbb{R}^{n+m}$. For $\|V\|$ -a.e. x , write $T_x V = T_x M$ and let

$$P_x : \mathbb{R}^{n+m} \rightarrow T_x V, \quad P_x^\perp := I - P_x$$

be the orthogonal projections.

Theorem 4.7 (Monotonicity formula). *Let V be a stationary integral n -varifold in $U \subset \mathbb{R}^{n+m}$. If*

$$B_s(x_0) \subset B_r(x_0) \subset\subset U,$$

then

$$\frac{\|V\|(B_r(x_0))}{\omega_n r^n} - \frac{\|V\|(B_s(x_0))}{\omega_n s^n} = \frac{1}{\omega_n} \int_{B_r(x_0) \setminus \overline{B}_s(x_0)} \frac{|P_x^\perp(x - x_0)|^2}{|x - x_0|^{n+2}} d\|V\|(x).$$

Here $\omega_n = \mathcal{L}^n(B_1^n)$.

Define the density ratio

$$\Theta_V(x_0, r) := \frac{\|V\|(B_r(x_0))}{\omega_n r^n}.$$

The formula immediately gives the following observations.

- (1) The function $r \mapsto \Theta_V(x_0, r)$ is monotone nondecreasing as long as $B_r(x_0) \subset\subset U$.
- (2) The density

$$\Theta_V(x_0) := \lim_{r \downarrow 0} \Theta_V(x_0, r)$$

exists at every point $x_0 \in U$.

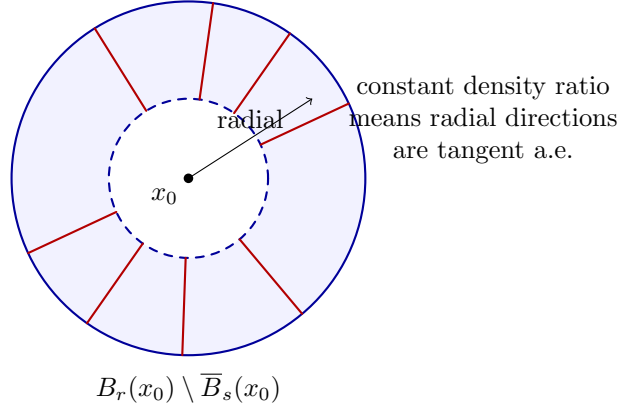
- (3) The function $\Theta_V(x_0, \cdot)$ is constant on $[s, r]$ if and only if

$$P_x^\perp(x - x_0) = 0 \quad \text{for } \|V\|\text{-a.e. } x \in B_r(x_0) \setminus \overline{B}_s(x_0),$$

equivalently

$$x - x_0 \in T_x V \quad \text{for } \|V\|\text{-a.e. } x \text{ in the annulus.}$$

In that case the radial vector field is tangent to V a.e. in the annulus, so the varifold is conical there: dilations centered at x_0 preserve the support and multiplicity as long as the dilated points remain in the annulus.



Proof of the monotonicity formula. By translating, assume $x_0 = 0$. Write $\rho(x) = |x|$ and let P_x denote projection onto $T_x V$. For a radial test vector field

$$X(x) = \gamma(\rho)x, \quad \gamma \in C_c^1((0, \infty)),$$

stationarity gives

$$0 = \delta V(X) = \int \operatorname{div}_{T_x V} X \, d\|V\|(x).$$

At a point where $T_x V$ exists,

$$\operatorname{div}_{T_x V}(\gamma(\rho)x) = n\gamma(\rho) + \rho\gamma'(\rho) \left| P_x \left(\frac{x}{\rho} \right) \right|^2.$$

Since

$$\left| P_x \left(\frac{x}{\rho} \right) \right|^2 = 1 - \left| P_x^\perp \left(\frac{x}{\rho} \right) \right|^2,$$

we obtain

$$\int (n\gamma(\rho) + \rho\gamma'(\rho)) \, d\|V\| = \int \rho\gamma'(\rho) \left| P_x^\perp \left(\frac{x}{\rho} \right) \right|^2 \, d\|V\|.$$

Now approximate the piecewise C^1 function

$$\gamma(\rho) = \begin{cases} s^{-n} - r^{-n}, & 0 < \rho \leq s, \\ \rho^{-n} - r^{-n}, & s < \rho < r, \\ 0, & \rho \geq r, \end{cases}$$

by smooth compactly supported radial cutoffs. Passing to the limit in the preceding identity gives

$$ns^{-n}\|V\|(B_s) - nr^{-n}\|V\|(B_r) = -n \int_{B_r \setminus \overline{B}_s} \frac{|P_x^\perp x|^2}{|x|^{n+2}} d\|V\|(x).$$

Dividing by $-n\omega_n$ yields

$$\frac{\|V\|(B_r)}{\omega_n r^n} - \frac{\|V\|(B_s)}{\omega_n s^n} = \frac{1}{\omega_n} \int_{B_r \setminus \overline{B}_s} \frac{|P_x^\perp x|^2}{|x|^{n+2}} d\|V\|(x),$$

which is the desired formula after translating back to the center x_0 . $\square$

Proposition 4.8 (Consequences of monotonicity). *Let $V = \underline{v}(M, \theta)$ be a stationary integral n -varifold in $U \subset \mathbb{R}^{n+m}$, and write*

$$\text{supp } V := \text{supp } \|V\|.$$

The following facts will be used repeatedly.

- (1) *For every $x \in U$, the density*

$$\Theta_V(x) := \lim_{r \downarrow 0} \Theta_V(x, r) = \inf_{B_r(x) \subset \subset U} \Theta_V(x, r)$$

exists. If $U = \mathbb{R}^{n+m}$, we also define the density at infinity by

$$\Theta_V(\infty) := \lim_{r \rightarrow \infty} \Theta_V(x, r) \in [0, \infty],$$

and this number is independent of the center x .

- (2) *For $\|V\|$ -a.e. x ,*

$$\Theta_V(x) = \theta(x).$$

Thus the density is the multiplicity at almost every ordinary point of the rectifiable support.

- (3) *The function $x \mapsto \Theta_V(x)$ is upper semicontinuous in U .*

- (4) *One has the support characterization*

$$\text{supp } V = \text{supp } \|V\| = \{x \in U : \Theta_V(x) \geq 1\}.$$

Consequently the support of a stationary integral varifold cannot be an arbitrary huge closure of a rectifiable set: it is precisely the positive density set, and it carries an n -dimensional lower density bound at each of its points. This rules out the kind of artificial example obtained by taking many tiny translated spheres with a very large closure.

- (5) *As a varifold, changing the multiplicity on an $\mathcal{H}^n$ -null set is invisible, so we may write*

$$V = \underline{v}(\text{supp } V, \Theta_V).$$

This is a useful mental picture: $\text{supp } V$ looks n -dimensional, while Θ_V records how many sheets are present. At singular points the density need not be an integer; for example the planar Y cone has density $\frac{3}{2}$ at the junction, although this point is $\mathcal{H}^1$ -null.

Proof. Only a few points need explanation. For the density at infinity, if $x, y \in \mathbb{R}^{n+m}$ and $d = |x - y|$, then for $r > d$,

$$B_{r-d}(y) \subset B_r(x) \subset B_{r+d}(y).$$

After dividing by $\omega_n r^n$ and using monotonicity for the center y , the factors $\left(\frac{r+d}{r}\right)^n$ tend to 1, so the two limits at infinity agree. The identity $\Theta_V = \theta$ for $\|V\|$ -a.e. x is the usual density theorem for rectifiable varifolds:

$$\|V\| = \theta \mathcal{H}^n \llcorner M.$$

For upper semicontinuity, let $x_j \rightarrow x$. Given $\rho > 0$ such that $B_\rho(x) \subset\subset U$, for every small $\varepsilon > 0$ and large j ,

$$B_\rho(x_j) \subset B_{\rho+\varepsilon}(x).$$

Since $\Theta_V(x_j) \leq \Theta_V(x_j, \rho)$, we get

$$\limsup_{j \rightarrow \infty} \Theta_V(x_j) \leq \frac{\|V\|(B_{\rho+\varepsilon}(x))}{\omega_n \rho^n}.$$

Letting $\varepsilon \downarrow 0$ along radii with $\|V\|(\partial B_{\rho+\varepsilon}(x)) = 0$, and then letting $\rho \downarrow 0$, gives $\limsup_j \Theta_V(x_j) \leq \Theta_V(x)$.

It remains to justify the support characterization. If $x \notin \text{supp } \|V\|$, then $\Theta_V(x) = 0$. Conversely, suppose $x \in \text{supp } \|V\|$. Fix $r > 0$ with $B_r(x) \subset\subset U$ and choose $\varepsilon > 0$. Since x is in the support, $B_{\varepsilon r}(x)$ has positive $\|V\|$ -mass, so it contains a $\|V\|$ -density point y of M with $\theta(y) \geq 1$. By monotonicity centered at y ,

$$\|V\|(B_r(x)) \geq \|V\|(B_{(1-\varepsilon)r}(y)) \geq \omega_n(1-\varepsilon)^n r^n.$$

Thus $\Theta_V(x, r) \geq (1-\varepsilon)^n$; sending $\varepsilon \downarrow 0$ and then $r \downarrow 0$ gives $\Theta_V(x) \geq 1$. $\square$

Proposition 4.9 (Local Hausdorff convergence of supports). *Let V_i be stationary integral n -varifolds in U , and suppose*

$$V_i \rightharpoonup V$$

as varifolds. Since the convergence gives local mass bounds and stationarity gives $|\delta V_i| = 0$, Theorem 2.26 implies that V is again integral; stationarity passes to the limit by testing against $\text{divs } X$. Moreover $\text{supp } V_i$ converges locally to $\text{supp } V$ in the Hausdorff sense: for every compact $K \subset\subset U$ and every $\varepsilon > 0$ with the ε -neighborhood of K still contained in U , for all large i ,

$$\text{supp } V_i \cap K \subset \{x : \text{dist}(x, \text{supp } V) < \varepsilon\},$$

and every point of $\text{supp } V \cap K$ lies within distance ε of $\text{supp } V_i$.

Proof. The second assertion follows directly from weak convergence of the mass measures: if $x \in \text{supp } V$ and $B_\varepsilon(x) \subset\subset U$, then $\|V\|(B_\varepsilon(x)) > 0$, hence $\|V_i\|(B_\varepsilon(x)) > 0$ for all large i . For the first assertion, argue by contradiction. Suppose $x_i \in \text{supp } V_i \cap K$, $x_i \rightarrow x \in K$, and $\text{dist}(x, \text{supp } V) \geq \varepsilon$. Then some ball $B_\rho(x) \subset\subset U \setminus \text{supp } V$ contains x_i for large i . By the previous proposition applied to V_i , every support point has density at least 1, so monotonicity gives a uniform lower bound

$$\|V_i\|(B_\rho(x)) \geq c(n, \rho) > 0$$

for large i . But $B_\rho(x)$ has zero $\|V\|$ -mass, contradicting the weak convergence after shrinking ρ slightly so that $\|V\|(\partial B_\rho(x)) = 0$. $\square$

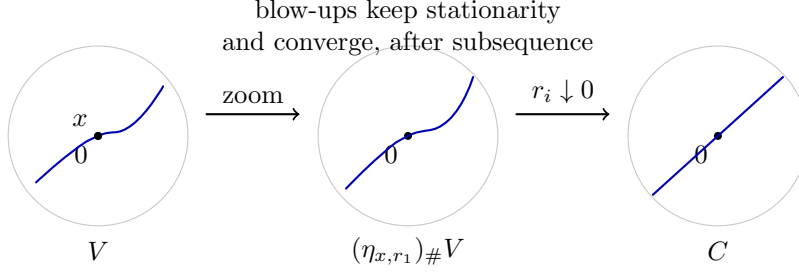
Definition 4.10. Let V be a stationary integral n -varifold in U and let $x \in \text{supp } V$. For $r > 0$ with $B_r(x) \subset\subset U$, set

$$\eta_{x,r}(y) := \frac{y-x}{r}, \quad V_{x,r} := (\eta_{x,r})_\# V.$$

A varifold C in $\mathbb{R}^{n+m}$ is called a tangent cone of V at x if there is a sequence $r_i \downarrow 0$ such that

$$V_{x,r_i} \rightharpoonup C$$

as varifolds on every compact subset of $\mathbb{R}^{n+m}$.



Proposition 4.11 (Existence and conicality of tangent cones). *Let V be a stationary integral n -varifold in U and let $x \in \text{supp } V$. For every sequence $r_i \downarrow 0$, a subsequence of V_{x,r_i} converges to a stationary integral n -varifold C in $\mathbb{R}^{n+m}$. Moreover C is dilation invariant:*

$$(\eta_{0,\lambda})\#C = C \quad \text{for every } \lambda > 0.$$

Thus tangent objects to stationary integral varifolds are stationary integral cones.

Proof. Fix $R < \infty$. For i large enough, $B_{Rr_i}(x) \subset\subset U$, and

$$\|V_{x,r_i}\|(B_R(0)) = r_i^{-n} \|V\|(B_{Rr_i}(x)) = \omega_n R^n \Theta_V(x, Rr_i).$$

The right side is uniformly bounded for large i because $\Theta_V(x, \rho)$ is monotone and $\rho = Rr_i$ is small. For each i , the map η_{x,r_i} is a smooth dilation and translation, so it sends

$$V = \underline{v}(M, \theta) \quad \text{to} \quad V_{x,r_i} = \underline{v}(\eta_{x,r_i}(M), \theta \circ \eta_{x,r_i}^{-1});$$

in particular rectifiability and integer multiplicity are preserved. Stationarity is also preserved: if $X \in C_c^1(\eta_{x,r_i}(U); \mathbb{R}^{n+m})$, then the pullback vector field

$$Y(y) := D\eta_{x,r_i}^{-1}(\eta_{x,r_i}(y)) X(\eta_{x,r_i}(y))$$

is compactly supported in U , and for this homothety $\delta V_{x,r_i}(X)$ is a positive constant multiple of $\delta V(Y)$; hence $\delta V_{x,r_i}(X) = 0$. Thus on every compact ball in $\mathbb{R}^{n+m}$ the sequence V_{x,r_i} has uniformly bounded mass and uniformly bounded first variation, in fact $|\delta V_{x,r_i}| = 0$. By Theorem 2.26, after passing to a subsequence there is a varifold limit C , and the same theorem is exactly what guarantees that C is still an integral n -varifold. The identity above also shows stationarity of C by passing to the limit in $\delta V_{x,r_i}(X) = 0$.

For every $0 < s < R$,

$$\Theta_{V_{x,r_i}}(0, R) - \Theta_{V_{x,r_i}}(0, s) = \Theta_V(x, Rr_i) - \Theta_V(x, sr_i) \rightarrow 0.$$

Passing to the limit gives $\Theta_C(0, R) = \Theta_C(0, s)$. By the equality case in the monotonicity formula,

$$y \in T_y C \quad \text{for } \|C\| \text{-a.e. } y \in B_R(0) \setminus \overline{B}_s(0).$$

Since s and R are arbitrary, the radial vector field is tangent to C a.e. away from the origin, which is exactly dilation invariance. $\square$

Remark 4.12. Tangent cones are the first models for singularities: after zooming in, the surface stops seeing lower order geometry and only the conical part remains. They need not be unique for general stationary varifolds. Kolář constructed rectifiable stationary 2-varifolds in $\mathbb{R}^4$ with non-unique conical tangent varifolds, and even examples with non-conical tangent varifolds [9]. In special minimizing or low-dimensional settings uniqueness can hold, but it is not a formal consequence of monotonicity alone.

Example 4.13 (Smooth submanifolds). Let $M^n \subset \mathbb{R}^{n+m}$ be a smooth embedded submanifold and let $V = |M|$. At every $x \in M$, the tangent cone is unique and equals

$$\text{Tan}(V, x) = |T_x M|.$$

Indeed, after translating x to the origin, M is the graph of a smooth function over $T_x M$ with value and first derivative zero at 0. Under the rescaling $y \mapsto \frac{y}{r}$, the higher order terms disappear, so the rescaled varifolds converge to the multiplicity-one tangent plane.

Example 4.14 (Triple junctions and the Y cone). Suppose three smooth sheets meet along a smooth $(n-1)$ -dimensional edge with equal surface tension and with 120° angles in the normal two-plane. At a point x on the edge, the blow-up freezes the edge to $T_x \Gamma \cong \mathbb{R}^{n-1}$ and freezes the normal cross-section to three half-lines meeting at 120° . Thus the tangent cone is

$$Y \times \mathbb{R}^{n-1},$$

where $Y \subset \mathbb{R}^2$ is the planar union of three rays meeting at 120° . In the surface case in $\mathbb{R}^3$, this is the usual soap-film Y cone: three half-planes meeting along one line.

Example 4.15 (Almost every point of a rectifiable varifold). If $V = \underline{v}(M, \theta)$ is an integral varifold, then for $\|V\|$ -a.e. $x \in \text{supp } \|V\|$ the approximate tangent plane $T_x M$ exists and

$$\text{Tan}(V, x) = \theta(x)|T_x M|.$$

This is the blow-up form of rectifiability: at almost every point, the set becomes flatter and flatter, while the integer multiplicity converges to the density $\theta(x)$.

Example 4.16 (Cones over smooth links). Let $\Sigma^{n-1} \subset \mathbb{S}^{n+m-1}$ be a smooth closed submanifold and define the cone

$$C(\Sigma) := \{r\xi : r > 0, \xi \in \Sigma\} \subset \mathbb{R}^{n+m}.$$

It is smooth away from the vertex. The correct statement is

$$C(\Sigma) \text{ is minimal in } \mathbb{R}^{n+m} \setminus \{0\} \iff \Sigma \text{ is minimal in } \mathbb{S}^{n+m-1}.$$

To see this, write the parametrization as $F(r, p) = r\phi(p)$, where $\phi : \Sigma \rightarrow \mathbb{S}^{n+m-1}$ is the inclusion. The cone metric is

$$g_C = dr^2 + r^2 g_\Sigma,$$

and the coordinate vector $x = F(r, p) = r\phi(p)$ satisfies

$$\Delta_C x = \frac{1}{r}(\Delta_\Sigma \phi + (n-1)\phi).$$

Since a submanifold of Euclidean space is minimal if and only if $\Delta x = 0$, the cone is minimal exactly when

$$\Delta_\Sigma \phi + (n-1)\phi = 0,$$

which is the minimal submanifold equation for $\Sigma \subset \mathbb{S}^{n+m-1}$. If Σ is closed, the same cone defines a stationary varifold in all of $\mathbb{R}^{n+m}$: the possible contribution from the vertex disappears by cutting out $B_\rho(0)$ and letting $\rho \downarrow 0$.

Example 4.17 (Two-dimensional cones and geodesic nets). When $n = 2$, a cone $C^2 \subset \mathbb{R}^{2+m}$ has a one-dimensional link

$$\Gamma = C \cap \mathbb{S}^{1+m}.$$

The preceding example says that smooth pieces of Γ must be geodesics in the sphere. If the cone has junctions, the link is not a smooth closed curve but a stationary integral 1-varifold in the sphere, called a geodesic net. Thus it consists of geodesic arcs with integer weights, and at every vertex the outgoing unit tangent vectors satisfy the balancing law

$$\sum_j q_j \tau_j = 0,$$

where $q_j \in \mathbb{N}$ are the weights. Conversely, the cone over any stationary geodesic net in the sphere is a stationary integral 2-cone. The Y cone is the simplest mental model: its link is a net of great-circle arcs meeting with the same 120° balance.

Example 4.18 (Clifford cones and the Simons cone). More generally, for $p, q \geq 1$, the product

$$\Sigma_{p,q} := \mathbb{S}^p \left(\sqrt{\frac{p}{p+q}} \right) \times \mathbb{S}^q \left(\sqrt{\frac{q}{p+q}} \right) \subset \mathbb{S}^{p+q+1}$$

is a minimal hypersurface of the unit sphere. Therefore $C(\Sigma_{p,q}) \subset \mathbb{R}^{p+q+2}$ is a minimal hypersurface cone. In the symmetric case $p = q = k$ this becomes

$$\mathbb{S}^k \left(\frac{1}{\sqrt{2}} \right) \times \mathbb{S}^k \left(\frac{1}{\sqrt{2}} \right) \subset \mathbb{S}^{2k+1}.$$

For $k = 3$, the cone is exactly the Simons cone

$$\{(x, y) \in \mathbb{R}^4 \times \mathbb{R}^4 : |x| = |y|\}.$$

This example is important because it is a genuine stationary minimal cone with an isolated singularity at the origin, and in the case $k = 3$ it is area-minimizing.

Theorem 4.19 (Allard regularity theorem, stationary form [1, 12]). *For every n, m and every $\alpha \in (0, 1)$ there are constants $\varepsilon = \varepsilon(n, m, \alpha) > 0$ and $C = C(n, m, \alpha)$ with the following property. Let V be a stationary integral n -varifold in $B_1^{n+m}(0)$ and assume $0 \in \text{supp } V$. Suppose that V is close to a multiplicity-one n -plane in either of the following two senses.*

- (1) *There is an n -plane $P \subset \mathbb{R}^{n+m}$ through the origin such that*

$$E(V, P, 1) := \int_{B_1(0)} \text{dist}(y, P)^2 d\|V\|(y) < \varepsilon^2, \quad \Theta_V(0, 1) \leq \frac{3}{2}.$$

- (2) *The density ratio is already close to one:*

$$\Theta_V(0, 1) = \frac{\|V\|(B_1(0))}{\omega_n} \leq 1 + \varepsilon.$$

Then there is an n -plane P_0 and a function

$$u : P_0 \cap B_{\frac{1}{2}}(0) \rightarrow P_0^\perp$$

such that

$$\text{supp } V \cap B_{\frac{1}{2}}(0) = \{p + u(p) : p \in P_0 \cap B_{\frac{1}{2}}(0)\}.$$

Moreover $u \in C^{1,\alpha}$ with quantitative estimates controlled by the small excess; in the first case, P_0 is close to the plane P appearing in the hypothesis. Since V is stationary, the graph satisfies the minimal surface system, hence elliptic bootstrapping makes u smooth. In particular, V is multiplicity one in $B_{\frac{1}{2}}(0)$.

Remark 4.20 (Why the density condition is enough). The first condition is the geometric form: the varifold has small quadratic height-excess over one plane, and the mass bound

$$\Theta_V(0, 1) \leq \frac{3}{2}$$

rules out two sheets over the same plane. The second condition is often more convenient.

The intuition is by contradiction. If a sequence had $\Theta_{V_i}(0, 1) \downarrow 1$ but were not regular, compactness would produce a stationary integral limit with density one at the origin. Monotonicity then forces the density ratio of the limit to be constant, hence the limit is a cone. An integral stationary cone of density one is a multiplicity-one plane. Thus, for large i , the varifold is close to a multiplicity-one plane and the first condition applies. Without smallness near density one, cones such as Y or the Simons cone may appear and regularity is false.

Example 4.21 (Why the hypotheses are necessary). The catenoid shows why one must exclude multiplicity. If a family of catenoids collapses onto a plane, then away from the neck the surfaces look like two sheets over the same plane and the varifold limit is $2|P|$. The height over P may be small, but the density is close to 2, not less than $\frac{3}{2}$; Allard's multiplicity-one theorem is not designed to regularize this into one graph.

The complex cusp

$$\mathcal{C} = \{(z, w) \in \mathbb{C}^2 : w^3 = z^2\}$$

is another obstruction. At the origin its tangent cone is the complex line $\{z = 0\}$ with multiplicity 2, because the parametrization $z = t^3$, $w = t^2$ has leading term $w = t^2$. Hence $\Theta_{\mathcal{C}}(0) = 2$. The set is area-minimizing as a complex analytic curve, but it is not a smooth embedded multiplicity-one graph at the origin; the density condition fails exactly where the singularity occurs.

Proposition 4.22 (Multiplicity-one tangent cone implies regularity). *Let V be a stationary integral n -varifold in U , and let $x \in \text{supp } V$. If one tangent cone of V at x is a multiplicity-one plane $|P|$, then V is a smooth embedded multiplicity-one minimal submanifold in a neighborhood of x .*

Proof. The tangent cone $|P|$ has density 1 at the origin. Since density is preserved under blow-up limits,

$$\Theta_V(x) = 1.$$

By monotonicity, $\Theta_V(x, r) \downarrow 1$ as $r \downarrow 0$. Choose $r > 0$ so small that $B_r(x) \subset\subset U$ and $\Theta_V(x, r) \leq 1 + \varepsilon$, where ε is the constant in Theorem 4.19. Apply the theorem to the rescaled varifold $(\eta_{x,r})_{\#} V$ in B_1 . This gives a $C^{1,\alpha}$ graph in $B_{\frac{r}{2}}(x)$; stationarity then upgrades it to a smooth minimal graph. $\square$

Proposition 4.23 (Regularity is stable under varifold convergence). *Let V_i be stationary integral n -varifolds in U with*

$$V_i \rightharpoonup V$$

as varifolds. Suppose $x_i \in \text{supp } V_i$, $x_i \rightarrow x \in \text{supp } V$, and that V is a smooth multiplicity-one varifold in a neighborhood of x . Then, for all sufficiently large i , V_i is a smooth multiplicity-one minimal submanifold in a neighborhood of x_i .

Proof. Choose $\rho > 0$ so small that $V = |M|$ in $B_{4\rho}(x)$, where M is a smooth embedded minimal graph over $T_x M$, and so that M has very small height-excess over $T_x M$ in $B_{4\rho}(x)$. Also choose ρ with

$$\|V\|(\partial B_{2\rho}(x)) = 0.$$

Since $V_i \rightharpoonup V$ and $x_i \rightarrow x$, for large i the rescaled varifolds centered at x_i have mass ratio less than $\frac{3}{2}$ in $B_{2\rho}(x_i)$ and have arbitrarily small height-excess over a plane close to $T_x M$.

Thus the first condition of Theorem 4.19 applies to V_i after scaling $B_{2\rho}(x_i)$ to B_1 . Therefore V_i is a $C^{1,\alpha}$ graph in $B_{\rho}(x_i)$, and stationarity again gives smoothness. $\square$

Remark 4.24 (Idea of the proof of Allard regularity). The proof is an elliptic regularity argument hidden inside the weak language of varifolds.

- (1) *Lipschitz approximation.* If V has small excess over $P = \mathbb{R}^n \times \{0\}$ and density close to one, most of $\text{supp } V$ can be written as the graph of a Lipschitz function $u : \mathbb{R}^n \rightarrow \mathbb{R}^m$. This is the step that converts measure-theoretic closeness to a plane into an approximate single-valued graph.
- (2) *Approximate equation.* A true minimal graph satisfies

$$g^{ij} D_{ij} u^\alpha = 0, \quad \alpha = 1, \dots, m,$$

where $g_{ij} = \delta_{ij} + D_i u \cdot D_j u$. If the slope is small, then $g^{ij} = \delta^{ij} + O(|Du|^2)$ and symbolically

$$\Delta u^\alpha = -(g^{ij} - \delta^{ij}) D_{ij} u^\alpha \sim Du * Du * D^2 u.$$

For a varifold one obtains this equation only weakly and approximately, but it is close enough to the harmonic equation.

- (3) *Harmonic approximation.* The approximate graph u is close in L^2 to a harmonic function v on the base plane. Harmonic functions have strong decay of their quadratic oscillation: after subtracting a best affine function,

$$\int_{B_r} |v - \ell_r|^2 \leq Cr^{n+2+2\alpha} \int_{B_1} |v - \ell_1|^2.$$

- (4) *Excess decay.* Transferring the harmonic decay back to the varifold gives the key estimate: for each $x \in \text{supp } V$ and each small r , there is an n -plane $P_x(r)$ such that

$$r^{-n-2} \int_{B_r(x)} \text{dist}(y, x + P_x(r))^2 d\|V\|(y) \leq Cr^{2\alpha}.$$

This says that the best tangent planes improve at smaller scales at a Hölder rate.

- (5) *From decay to a graph.* The excess decay implies that tangent planes vary Hölder continuously, so $\text{supp } V$ is a $C^{1,\alpha}$ graph. Once the graph structure is known, the stationary equation becomes the classical minimal surface system, and standard elliptic bootstrapping gives smoothness.

Definition 4.25. Let V be an integral n -varifold in U . A point $x \in \text{supp } V$ is called a regular point if there are $\rho > 0$, a smooth embedded n -submanifold $M \subset B_\rho(x)$, and an integer $q \geq 1$ such that

$$V \llcorner B_\rho(x) = q|M|.$$

We write the set of such points as $\text{reg } V$, and set

$$\text{sing } V := \text{supp } V \setminus \text{reg } V.$$

Proposition 4.26 (First applications of Allard regularity). *Let V be a stationary integral n -varifold in U .*

- (1) *The set $\text{reg } V$ is relatively open and dense in $\text{supp } V$.*
- (2) *If V has multiplicity one, in the sense that $V = \underline{v}(\text{supp } V, 1)$ up to $\mathcal{H}^n$ -null sets, then*

$$\mathcal{H}^n(\text{sing } V) = 0.$$

Proof. The set $\text{reg } V$ is relatively open by definition and by smoothness of the local graphical representation. To see density, take any ball $B_\rho(x) \subset U$ meeting $\text{supp } V$. Since $\|V\|(B_\rho(x)) > 0$, rectifiability gives a point $y \in B_\rho(x)$ at which the approximate tangent plane exists and

$$\text{Tan}(V, y) = q|P|$$

for some integer $q \geq 1$ and some n -plane P . The integer-multiplicity version of Allard's theorem says that if, after rescaling, a stationary integral varifold is sufficiently close to $q|P|$ and its density is less than $q + \frac{1}{2}$, then it is a smooth graph with constant multiplicity q . Applying this at a sufficiently small scale around y shows $y \in \text{reg } V$. Hence every support ball contains a regular point.

If V has multiplicity one, then for $\|V\|$ -a.e. y the tangent cone is $|T_y M|$. By Proposition 4.22, such points are regular. Therefore $\|V\|(\text{sing } V) = 0$. Since in the multiplicity-one case $\|V\| = \mathcal{H}^n \llcorner \text{supp } V$, this is exactly $\mathcal{H}^n(\text{sing } V) = 0$. $\square$

Remark 4.27 (Why minimizers are better than stationary varifolds). Allard regularity reduces regularity to understanding tangent cones. For a general stationary varifold, non-flat stationary cones such as the Y cone or the Simons cone may occur. For mass-minimizing currents, the tangent cones are not merely stationary cones: they are minimizing cones. In dimensions where all minimizing cones of the relevant type are planes, Allard regularity then forces smoothness. This

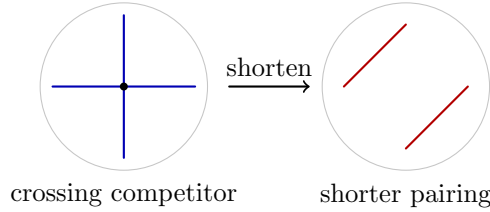
is the guiding strategy behind the regularity theory of minimizers: classify or exclude non-planar minimizing cones.

Definition 4.28. Let $T \in \mathcal{I}_n(U)$, where $U \subset \mathbb{R}^{n+m}$ is open. We say that T is mass-minimizing in U if, for every open set $W \subset\subset U$ and every $S \in \mathcal{I}_{n+1}(U)$ with $\text{supp } S \subset W$,

$$\|T\|(W) \leq \|T + \partial S\|(W).$$

Equivalently, no compactly supported integral filling ∂S inside U can reduce the mass of T .

Example 4.29 (One-dimensional minimizing currents). Any one-dimensional mass-minimizing integral current in a Euclidean open set is a locally finite union of oriented straight line segments, rays, and lines, with integer multiplicities. The pieces are disjoint in their interiors; endpoints can occur only on $\text{supp } \partial T$ or at the boundary of the open set under consideration.



Indeed, at an interior point of a minimizing current every tangent cone is a one-dimensional minimizing cone. Such a cone is a finite integer combination of rays from the origin:

$$C = \sum_j q_j \llbracket R_j \rrbracket.$$

If two rays are not opposite, replacing the broken path by the straight chord gives a shorter current with the same boundary on a small sphere. Thus the only boundary-free minimizing cone is an integer multiple of a line.

Allard regularity then makes the current a smooth curve near every interior point. Finally, a smooth minimizing curve in Euclidean space must be a straight geodesic; otherwise a short arc could be replaced by its chord. The same chord-shortening argument rules out interior intersections, because crossing or branching produces a strictly shorter pairing with the same boundary data.

Proposition 4.30 (Stationarity away from the boundary). *If T is mass-minimizing in U , then the associated varifold $|T|$ is stationary in*

$$U \setminus \text{supp } \partial T.$$

Proof. Let $X \in C_c^1(U \setminus \text{supp } \partial T; \mathbb{R}^{n+m})$ and let Φ_t be its flow. For $|t|$ small, $\Phi_{t\#}T - T$ is the boundary of an integral $(n+1)$ -current supported in a compact subset of $U \setminus \text{supp } \partial T$; this is the homotopy formula for currents. By minimality,

$$\mathbf{M}(\Phi_{t\#}T) \geq \mathbf{M}(T)$$

for both positive and negative t . Hence the derivative at $t = 0$ vanishes. The first variation of mass is the first variation of the associated varifold, so $\delta|T|(X) = 0$. $\square$

Theorem 4.31 (Compactness for minimizing currents). *Let $T_i \in \mathcal{I}_n(U)$ be mass-minimizing in U . Suppose that for every $W \subset\subset U$,*

$$\sup_i (\|T_i\|(W) + \|\partial T_i\|(W)) < \infty.$$

Then, after passing to a subsequence, there is an integral current $T \in \mathcal{I}_n(U)$ such that

$$T_i \rightharpoonup T$$

as currents, and T is again mass-minimizing in U . Moreover, on compact subsets of $U \setminus \text{supp } \partial T$,

$$|T_i| \rightharpoonup |T|$$

as varifolds and the supports $\text{supp } T_i$ converge locally to $\text{supp } T$ in Hausdorff distance. If $\partial T_i = 0$ in U , then these varifold and support convergences hold locally throughout U .

Proof sketch. Federer–Fleming compactness gives a current subsequential limit T . The class of mass-minimizing currents is closed under current convergence: one tests against a fixed compactly supported competitor, uses the minimizing inequality for T_i , and passes to the limit using the standard lower semicontinuity and deformation argument from the direct method.

Away from the limiting boundary the associated varifolds $|T_i|$ are stationary and have locally uniformly bounded mass, so Allard compactness gives varifold convergence. No cancellation occurs for minimizing currents in the interior, hence the varifold limit is $|T|$. The local Hausdorff convergence of supports then follows from the density lower bound for stationary integral varifolds, as in Proposition 4.9. $\square$

Definition 4.32. Let T be mass-minimizing in U and let

$$x \in \text{supp } T \setminus \text{supp } \partial T.$$

For $r > 0$ with $B_r(x) \subset\subset U \setminus \text{supp } \partial T$, define

$$T_{x,r} := (\eta_{x,r})_{\#} T, \quad \eta_{x,r}(y) = \frac{y - x}{r}.$$

A current T' in $\mathbb{R}^{n+m}$ is called a minimizing tangent cone of T at x if there is a sequence $r_i \downarrow 0$ such that

$$T_{x,r_i} \rightharpoonup T'$$

as currents on every compact subset of $\mathbb{R}^{n+m}$.

Proposition 4.33 (Existence of minimizing tangent cones). *Let T be mass-minimizing in U and let $x \in \text{supp } T \setminus \text{supp } \partial T$. For every sequence $r_i \downarrow 0$, a subsequence of T_{x,r_i} converges to an integral current T' in $\mathbb{R}^{n+m}$. The limit satisfies*

$$\partial T' = 0, \quad T' \text{ is mass-minimizing in } \mathbb{R}^{n+m},$$

and it is conical:

$$(\eta_{0,\lambda})_{\#} T' = T' \quad \text{for every } \lambda > 0.$$

Proof. Fix $R < \infty$. For i large enough, $B_{Rr_i}(x) \subset\subset U \setminus \text{supp } \partial T$, so $\partial T_{x,r_i} = 0$ in $B_R(0)$. The monotonicity formula for $|T|$ gives

$$\|T_{x,r_i}\|(B_R(0)) = r_i^{-n} \|T\|(B_{Rr_i}(x)) \leq C(R)$$

for all large i . Federer–Fleming compactness therefore gives a current limit on each B_R , and a diagonal argument gives a limit T' on all of $\mathbb{R}^{n+m}$. Since the rescaled currents are minimizing and have no boundary in each fixed ball for large i , the compactness theorem above implies that T' is minimizing and $\partial T' = 0$.

The associated varifolds $|T_{x,r_i}|$ are blow-ups of the stationary varifold $|T|$, hence their limits are stationary cones by the tangent-cone argument above. Since the compactness theorem gives $|T_{x,r_i}| \rightharpoonup |T'|$, the varifold $|T'|$ is dilation invariant. The current orientations and integer multiplicities are also preserved under dilation; equivalently, applying the equality case in the current monotonicity formula shows $(\eta_{0,\lambda})_{\#} T' = T'$. Thus T' is a minimizing integral conical n -current. $\square$

4.4. Codimension One and the Reduction to Minimizing Cones. The goal is now clear. If every minimizing tangent cone in the relevant dimension and codimension is a multiplicity-one plane, then the Allard consequence above implies regularity. In codimension one this goal can be reduced to the theory of finite-perimeter sets.

Proposition 4.34 (Local layer decomposition in codimension one). *Let $U \subset \mathbb{R}^{n+1}$ be open and let $T \in \mathcal{I}_n(U)$ satisfy $\partial T = 0$ in U . In every ball $B \subset\subset U$ there is an integer-valued function*

$$\theta \in BV(B; \mathbb{Z})$$

such that, if S_θ denotes the top-dimensional current

$$S_\theta(f \, dx^1 \wedge \cdots \wedge dx^{n+1}) = \int_B \theta f \, d\mathcal{L}^{n+1},$$

then

$$T \llcorner B = \partial S_\theta.$$

Writing

$$E_j^+ = \{\theta \geq j\}, \quad E_j^- = \{\theta \leq -j\}, \quad j = 1, 2, \dots,$$

we have nested finite-perimeter layers $E_{j+1}^+ \subset E_j^+$ and $E_{j+1}^- \subset E_j^-$, and

$$T \llcorner B = \sum_{j=1}^{\infty} \partial \llbracket E_j^+ \rrbracket - \sum_{j=1}^{\infty} \partial \llbracket E_j^- \rrbracket \quad \text{in } B.$$

Moreover the decomposition has no hidden cancellation:

$$\|T\|(B) = \sum_{j=1}^{\infty} \|\partial \llbracket E_j^+ \rrbracket\|(B) + \sum_{j=1}^{\infty} \|\partial \llbracket E_j^- \rrbracket\|(B).$$

Explanation. This is the local form of Proposition 3.21. All equalities are understood inside the open ball B , that is, after testing against forms compactly supported in B , so no boundary term from ∂B appears. Since B has no n -dimensional homology, the cycle $T \llcorner B$ is the boundary of a top-dimensional integral current in B . Top-dimensional currents in $B \subset \mathbb{R}^{n+1}$ are exactly integer-valued BV densities, so the filling is S_θ for some $\theta \in BV(B; \mathbb{Z})$. The layer-cake identity

$$\theta = \sum_{j=1}^{\infty} \chi_{E_j^+} - \sum_{j=1}^{\infty} \chi_{E_j^-}$$

and the BV coarea formula give both the current identity and the mass identity. $\square$

Remark 4.35 (Why this is codimension one). The statement uses that a codimension-one cycle in $\mathbb{R}^{n+1}$ is the boundary of a top-dimensional integer-valued BV function. There is no analogous reduction in codimension two. For instance the complex curve

$$\{(z, w) \in \mathbb{C}^2 : w^3 = z^2\}$$

is an area-minimizing real 2-current in $\mathbb{R}^4$ with an isolated branch singularity; it is not the boundary of a finite-perimeter region in the ambient space. This is one reason the higher-codimension theory allows singularities already in dimension 2.

Proposition 4.36 (Each layer is minimizing). *Assume T is mass-minimizing in U and use the layer decomposition above in a ball $B \subset\subset U$. Then each oriented layer $\partial \llbracket E_j^+ \rrbracket$ and $-\partial \llbracket E_j^- \rrbracket$ is itself mass-minimizing in B .*

Proof. Let T_j be one of the oriented layers and write $T = T_j + R$ in B , with

$$\|T\|(B) = \|T_j\|(B) + \|R\|(B).$$

Suppose T_j were not minimizing. Then there would be an integral $(n+1)$ -current S with $\text{supp } S \subset W \subset\subset B$ such that

$$\|T_j + \partial S\|(W) < \|T_j\|(W).$$

Since $T + \partial S = R + (T_j + \partial S)$, the triangle inequality for mass gives

$$\|T + \partial S\|(W) \leq \|R\|(W) + \|T_j + \partial S\|(W) < \|R\|(W) + \|T_j\|(W) = \|T\|(W),$$

contradicting the minimizing property of T . Thus every layer is minimizing. $\square$

Remark 4.37 (Reduction of the regularity question). After separating the canonical layers and grouping coincident smooth layers into one integer-multiplicity sheet, one has the local equivalence

$$x \in \text{sing } T \iff x \in \text{sing}(\partial[E_j^+]) \text{ or } x \in \text{sing}(-\partial[E_j^-]) \text{ for some active layer } j.$$

The maximum principle prevents two distinct smooth minimizing hypersurface layers from crossing in the interior. Hence the essential local model is

$$T = \partial[E],$$

which is multiplicity one as a current and whose associated varifold has density one at almost every reduced-boundary point.

Theorem 4.38 (Simons' cone theorem). *Let $C = \partial[E]$ be a mass-minimizing conical integral n -current in $\mathbb{R}^{n+1}$. Assume that C is smooth away from the origin. If $1 \leq n \leq 6$, then*

$$C = \pm[P]$$

for some n -plane P through the origin. The result is sharp: in dimension $n = 7$ the Simons cone

$$\{(x, y) \in \mathbb{R}^4 \times \mathbb{R}^4 : |x| = |y|\}$$

is area-minimizing and singular at the origin [13, 3].

Remark 4.39 (Idea of Simons' argument). For $n = 1$ the statement is the one-dimensional classification already discussed: a minimizing cone is a line. Now assume $2 \leq n \leq 6$ and write the cone as

$$C = C(\Sigma) = \{r\omega : r > 0, \omega \in \Sigma\},$$

where $\Sigma^{n-1} \subset \mathbb{S}^n$ is the smooth link. Since C is minimizing, its smooth part is stable. For a two-sided smooth minimal hypersurface $M \subset \mathbb{R}^{n+1}$, stability says

$$\int_M |A_M|^2 \phi^2 d\mathcal{H}^n \leq \int_M |\nabla_M \phi|^2 d\mathcal{H}^n \quad \text{for all } \phi \in C_c^1(M).$$

For a cone, $|A_C|^2 = r^{-2}|A_\Sigma|^2$ and the cone metric is $dr^2 + r^2 g_\Sigma$. Testing the stability inequality with separated functions $\phi(r, \omega) = \zeta(r)\psi(\omega)$ and using the sharp Hardy inequality

$$\int_0^\infty |\zeta'(r)|^2 r^{n-1} dr \geq \frac{(n-2)^2}{4} \int_0^\infty \zeta(r)^2 r^{n-3} dr$$

shows that the first eigenvalue

$$\lambda_1 := \inf_{\psi \neq 0} \frac{\int_\Sigma (|\nabla_\Sigma \psi|^2 - |A_\Sigma|^2 \psi^2) d\mathcal{H}^{n-1}}{\int_\Sigma \psi^2 d\mathcal{H}^{n-1}}$$

satisfies

$$\lambda_1 \geq -\frac{(n-2)^2}{4}.$$

On the other hand, Simons' identity for a minimal hypersurface $\Sigma^{n-1} \subset \mathbb{S}^n$ is

$$\frac{1}{2}\Delta_\Sigma |A_\Sigma|^2 = |\nabla_\Sigma A_\Sigma|^2 + (n-1)|A_\Sigma|^2 - |A_\Sigma|^4.$$

If $A_\Sigma \not\equiv 0$, integrate this identity over the compact link and test the eigenvalue quotient with $\psi = |A_\Sigma|$. Since $|\nabla |A_\Sigma||^2 \leq |\nabla A_\Sigma|^2$, one obtains

$$\lambda_1 \leq -(n-1).$$

Thus a non-flat stable cone would force

$$n-1 \leq \frac{(n-2)^2}{4}.$$

This inequality first holds at $n = 7$. Therefore for $2 \leq n \leq 6$ one must have $A_\Sigma \equiv 0$, so Σ is a totally geodesic equator and C is a plane.

Definition 4.40 (Spine of a minimizing cone). Let C be a mass-minimizing conical boundary current in $\mathbb{R}^{n+1}$ and write

$$\Theta_C(x) := \Theta_{|C|}(x).$$

The *spine* of C is

$$S(C) := \{z \in \mathbb{R}^{n+1} : \Theta_C(z) = \Theta_C(0)\}.$$

Equivalently, $S(C)$ is the largest linear subspace along which the associated varifold $|C|$ is translation invariant. This equivalence is a standard consequence of the equality case in the monotonicity formula.

Lemma 4.41 (Spine splitting). *Let C be a mass-minimizing conical boundary current in $\mathbb{R}^{n+1}$, and let $q = \dim S(C)$. After a rotation,*

$$C = [\mathbb{R}^q] \times C',$$

where C' is a mass-minimizing conical boundary current in $\mathbb{R}^{n+1-q}$ and $S(C') = \{0\}$. Moreover, if $y \in \text{sing } C \setminus S(C)$ and D is a tangent cone of C at y , then

$$\dim S(D) \geq q + 1.$$

Proof. The first statement is the translation-invariance interpretation of the spine. Invariance along $S(C)$ allows one to quotient out the Euclidean factor and obtain the residual cone C' in the orthogonal complement.

For the second statement, the blow-up of C at y still keeps the old translations along $S(C)$. In addition, because C is a cone with vertex 0, zooming at a point $y \notin S(C)$ creates translation invariance in the new direction represented by y modulo $S(C)$. Thus a tangent cone at y has at least one more line in its spine. If that tangent cone were a multiplicity-one plane, Allard regularity would imply that y was regular for C , contradicting $y \in \text{sing } C$. $\square$

Definition 4.42 (Simon's strata). Let $T = \partial[E]$ be mass-minimizing in $U \subset \mathbb{R}^{n+1}$. For $0 \leq q \leq n$, define

$$\mathcal{S}^q(T) := \{x \in \text{supp } T \cap U : \dim S(C) \leq q \text{ for every tangent cone } C \text{ of } T \text{ at } x\}.$$

Thus $\mathcal{S}^0(T)$ consists of points all of whose tangent cones have no line of translation invariance, while larger strata allow larger Euclidean factors. We always have

$$\mathcal{S}^0(T) \subset \mathcal{S}^1(T) \subset \cdots \subset \mathcal{S}^n(T).$$

Proposition 4.43 (Dimension estimate for the strata [12, Chapter 3]). *If $T = \partial[E]$ is mass-minimizing in $U \subset \mathbb{R}^{n+1}$, then*

$$\dim_{\mathcal{H}} \mathcal{S}^q(T) \leq q, \quad q = 0, \dots, n.$$

Idea of proof. This is the clean form of Federer's dimension-reduction argument. Suppose, to the contrary, that

$$\mathcal{H}^s(\mathcal{S}^q(T)) > 0 \quad \text{for some } s > q.$$

Choose an s -density point x of $\mathcal{S}^q(T)$ and blow up T at x . Compactness gives a tangent cone C_0 .

The density-point choice forces the rescaled sets $\mathcal{S}^q(T)$ to leave a nontrivial trace in the stratum $\mathcal{S}^q(C_0)$. Since $s > q$ and $\dim S(C_0) \leq q$, this trace cannot be contained in the spine $S(C_0)$. Hence one can choose a point

$$y_0 \in \mathcal{S}^q(C_0) \setminus S(C_0)$$

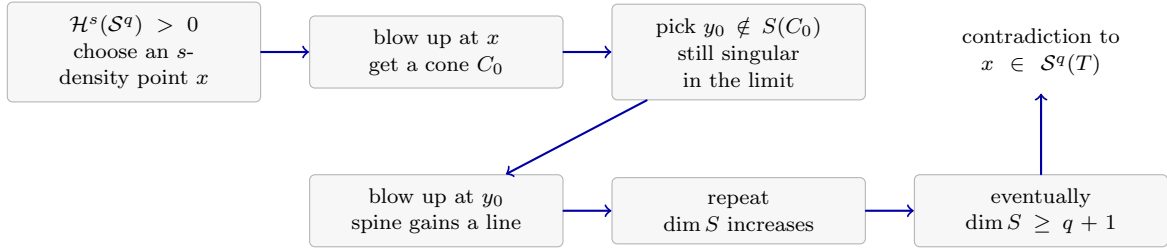
detected by the blown-up stratum. Such a point is singular, because a regular point has a tangent plane whose spine has dimension n . Blow up C_0 at y_0 . By Lemma 4.41, the new tangent cone C_1 has

$$\dim S(C_1) \geq \dim S(C_0) + 1.$$

Repeating the same density-point argument produces, after at most $q + 1$ steps, a tangent cone of T at x whose spine has dimension at least $q + 1$. This contradicts the defining property $x \in \mathcal{S}^q(T)$. Therefore

$$\mathcal{H}^s(\mathcal{S}^q(T)) = 0 \quad \text{for every } s > q,$$

which is exactly $\dim_{\mathcal{H}} \mathcal{S}^q(T) \leq q$. □



Theorem 4.44 (Federer dimension reduction principle). *Fix an integer $k \geq 2$. Assume that for every $1 \leq j < k$, every mass-minimizing conical boundary current*

$$C = \partial \llbracket E \rrbracket \quad \text{in } \mathbb{R}^{j+1}$$

with E of locally finite perimeter, which is smooth away from the origin, is a multiplicity-one plane. Let

$$T = \partial \llbracket E \rrbracket$$

be mass-minimizing in an open set $U \subset \mathbb{R}^{n+1}$, with E of locally finite perimeter. Then

$$\dim_{\mathcal{H}} \text{sing } T \leq n - k.$$

In the borderline case $n = k$, the singular set is discrete. If $n < k$, the singular set is empty.

Proof. We first show that

$$\text{sing } T \subset \mathcal{S}^{n-k}(T).$$

Indeed, suppose that $x \in \text{sing } T$ and that some tangent cone C of T at x has spine dimension $q \geq n - k + 1$. By Lemma 4.41,

$$C = \llbracket \mathbb{R}^q \rrbracket \times C',$$

where C' is a mass-minimizing conical boundary current of dimension $n - q < k$. Since x is singular, C is not a multiplicity-one plane, so the residual cone C' is not a multiplicity-one plane.

If C' is smooth away from the origin, this contradicts the hypothesis. If not, choose a nonzero singular point of C' and blow up there. Lemma 4.41 adds another line factor and lowers the residual dimension. Repeating this finite process gives a non-planar minimizing conical boundary current of some dimension

$$1 \leq j < k$$

which is smooth away from the origin, again contradicting the hypothesis. Hence no such tangent cone can occur, and the inclusion follows.

By Proposition 4.43,

$$\dim_{\mathcal{H}} \operatorname{sing} T \leq \dim_{\mathcal{H}} \mathcal{S}^{n-k}(T) \leq n - k.$$

This proves the Hausdorff dimension estimate.

When $n = k$, the estimate says that the singular set is zero-dimensional, but one also wants local discreteness. If singular points $x_i \rightarrow x \in U$ accumulated, then after blowing up T at x using scales comparable to $|x_i - x|$, one would obtain a tangent cone with a nonzero singular point. Blowing up that cone at the nonzero singular point gives a non-planar minimizing boundary cone of dimension $k - 1 < k$, and the same finite reduction as above contradicts the hypothesis. Thus $\operatorname{sing} T$ has no interior accumulation point and is discrete.

If $n < k$, any singular point would immediately yield a non-planar minimizing cone of dimension $n < k$ after the same reduction, so no singular point exists. $\square$

Corollary 4.45 (No non-planar minimizing boundary cones for $n \leq 6$). *Every mass-minimizing conical current of the form $\partial \llbracket E \rrbracket$ in $\mathbb{R}^{n+1}$, with $n \leq 6$, is a multiplicity-one plane.*

Idea of the reduction. Take $k = 7$ in Theorem 4.44. Simons' cone theorem verifies the hypothesis for every $j < 7$: in dimensions $2, \dots, 6$ a smooth-link minimizing cone must be a plane, and in dimension 1 the one-dimensional classification gives the same conclusion.

Therefore a minimizing boundary cone of dimension $n \leq 6$ has no singularity. A smooth cone through the origin must be equal to its tangent plane at the origin, hence it is a multiplicity-one plane. $\square$

Theorem 4.46 (Smoothness of multiplicity-one codimension-one minimizers). *Let $U \subset \mathbb{R}^{n+1}$ be open and let*

$$T = \partial \llbracket E \rrbracket$$

be mass-minimizing in U , where E has locally finite perimeter. If $n \leq 6$, then T is smooth in U : more precisely, $\operatorname{supp} T \cap U$ is a smooth embedded minimal hypersurface and

$$T = \pm \llbracket M \rrbracket$$

locally, with the sign determined by the orientation of E .

Proof. Let $x \in \operatorname{supp} T \cap U$. Since $\partial T = 0$ in U , every blow-up sequence has a minimizing tangent cone C at x . Because $T = \partial \llbracket E \rrbracket$, the blow-up is again a multiplicity-one minimizing boundary cone. By Corollary 4.45, C is a multiplicity-one plane.

Therefore some tangent cone of $|T|$ at x is a multiplicity-one plane. Proposition 4.22, the Allard consequence proved above, implies that $|T|$ is regular near x . Since x was arbitrary, T is smooth throughout U . $\square$

Theorem 4.47 (Codimension-one regularity theorem). *Let $U \subset \mathbb{R}^{n+1}$ be open and let $T \in \mathcal{I}_n(U)$ be mass-minimizing with*

$$\partial T = 0 \quad \text{in } U.$$

Then $\operatorname{reg} T$ is relatively open in $\operatorname{supp} T$, and near every regular point

$$T = q \llbracket M \rrbracket$$

for an integer $q \geq 1$ and a smooth embedded minimal hypersurface M . The singular set satisfies

$$\begin{cases} \operatorname{sing} T = \emptyset, & n \leq 6, \\ \operatorname{sing} T \text{ is discrete,} & n = 7, \\ \dim_{\mathcal{H}} \operatorname{sing} T \leq n - 7, & n \geq 8. \end{cases}$$

Proof. Work locally in a ball $B \subset\subset U$. By Proposition 4.34,

$$T \llcorner B = \sum_j \partial \llbracket E_j^+ \rrbracket - \sum_j \partial \llbracket E_j^- \rrbracket$$

with no cancellation in mass, and by Proposition 4.36 each layer is a mass-minimizing boundary current. Simons' theorem supplies the cone-classification hypothesis in dimensions < 7 , so Theorem 4.44 applied with $k = 7$ gives the estimates for the singular set of each layer. On compact subballs only finitely many nonzero layers can meet the subball: otherwise the monotonicity lower bound would force infinite mass. Thus the union of the layer singular sets has the same dimension bound, and in the discrete case remains locally finite.

Away from these layer singular sets, every active layer is a smooth embedded minimizing hypersurface. The maximum principle prevents two different smooth minimizing layers from crossing; if they touch on one side, they coincide locally. Therefore the smooth layers combine into locally constant integer multiplicity sheets, giving $T = q \llbracket M \rrbracket$ near each regular point. This proves the complete codimension-one interior regularity statement. $\square$

5. BERNSTEIN THEOREMS

5.1. The Bernstein Problem for Minimal Graphs. We now return to the word “Bernstein” in the title. The previous sections developed weak surfaces, compactness, tangent cones, and regularity. The Bernstein problem asks what those tools say in the most rigid global graphical situation. Throughout this chapter we keep the presentation short and use only the pieces of regularity theory already introduced; for a systematic treatment of the classical theory, see Colding–Minicozzi [5].

Let $u : \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$ be a C^2 function and consider its graph

$$M = \text{graph } u = \{(x, u(x)) : x \in \Omega\} \subset \mathbb{R}^{n+1}.$$

The induced metric on M is

$$g_{ij} = \delta_{ij} + D_i u D_j u,$$

so the area functional is

$$\mathbf{A}(u; \Omega) = \int_{\Omega} \sqrt{1 + |Du|^2} \, dx.$$

If $u_s = u + s\phi$ with $\phi \in C_c^\infty(\Omega)$, then

$$\left. \frac{d}{ds} \right|_{s=0} \mathbf{A}(u_s; \Omega) = \int_{\Omega} \frac{Du \cdot D\phi}{\sqrt{1 + |Du|^2}} \, dx = - \int_{\Omega} \operatorname{div} \left(\frac{Du}{\sqrt{1 + |Du|^2}} \right) \phi \, dx.$$

Thus M is stationary under compactly supported graphical variations if and only if

$$\operatorname{div} \left(\frac{Du}{\sqrt{1 + |Du|^2}} \right) = 0. \tag{MSE}$$

This is the scalar minimal surface equation. Equivalently, the upward unit normal

$$\nu = \frac{(-Du, 1)}{\sqrt{1 + |Du|^2}}$$

has zero tangential mean curvature on the graph, so $H_M = 0$.

Definition 5.1. An entire minimal graph in $\mathbb{R}^{n+1}$ is a graph u where

$$u : \mathbb{R}^n \rightarrow \mathbb{R}$$

is a smooth solution of the minimal surface equation. Affine functions

$$u(x) = a \cdot x + b$$

give the basic examples; their graphs are n -planes.

Question 5.2 (Bernstein problem). If $u : \mathbb{R}^n \rightarrow \mathbb{R}$ is an entire solution of the minimal surface equation, must u be affine?

In geometric language, the question asks whether every complete minimal hypersurface in $\mathbb{R}^{n+1}$ that is globally a single-valued graph over a plane must itself be a plane. This is much stronger than local regularity. Local regularity says that a minimizer or a stationary varifold is smooth once its tangent cone is a multiplicity-one plane. Bernstein asks whether a global graphical assumption forces flatness at every scale.

There is a useful bridge from the PDE to the GMT language. If u solves (MSE), define the subgraph

$$E_u = \{(x, t) \in \Omega \times \mathbb{R} : t < u(x)\}.$$

Extend the unit normal vertically by

$$X(x, t) = \frac{(-Du(x), 1)}{\sqrt{1 + |Du(x)|^2}}.$$

Then

$$\operatorname{div}_{\mathbb{R}^{n+1}} X = -\operatorname{div}_{\mathbb{R}^n} \left(\frac{Du}{\sqrt{1 + |Du|^2}} \right) = 0.$$

Hence the n -form $\omega = \iota_X(dx^1 \wedge \cdots \wedge dx^{n+1})$ is closed, has comass at most 1, and restricts to the volume form on graph u with the upward orientation. Thus ω calibrates the graph. In particular, $\partial[E_u]$ is locally mass-minimizing in cylinders contained in $\Omega \times \mathbb{R}$.

We shall use the same current and varifold notation as before:

$$T_u := \partial[E_u], \quad V_u := |T_u|.$$

Thus $\operatorname{supp} T_u = M$ and $\|T_u\| = \|V_u\| = \mathcal{H}^n \llcorner M$ when the graph is smooth and has multiplicity one.

For an entire graph, one can therefore combine two viewpoints. Analytically, (MSE) is a nonlinear elliptic equation. Geometrically, the graph is a complete locally minimizing hypersurface. If we blow it down,

$$T_{u,r} := (\eta_{0,r})_{\#} T_u, \quad V_{u,r} := |T_{u,r}|, \quad \eta_{0,r}(y) = \frac{y}{r},$$

then $\operatorname{supp} T_{u,r} = r^{-1}M$. Compactness gives subsequential limits that are minimizing cones. Thus Bernstein-type theorems are tied to the classification of minimizing cones: if the only possible blow-down cones are multiplicity-one planes, then the graph should be forced toward flatness. The next subsections make this principle precise in the classical two-dimensional theorem and then explain the higher-dimensional threshold.

5.2. The Classical Bernstein Theorem in $\mathbb{R}^3$. We now prove the original Bernstein theorem. The proof is one of the cleanest places where the complex analysis of minimal surfaces meets the global graphical assumption.

Theorem 5.3 (Bernstein theorem in dimension two). *Let $u : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a smooth entire solution of the minimal surface equation*

$$\operatorname{div} \left(\frac{Du}{\sqrt{1 + |Du|^2}} \right) = 0.$$

Then u is affine. Equivalently, every entire minimal graph in $\mathbb{R}^3$ is a plane.

We use two standard facts from the classical theory of minimal surfaces.

Lemma 5.4 (Conformal type of an entire minimal graph). *Let $M = \text{graph } u \subset \mathbb{R}^3$ be an entire minimal graph. Then M admits a global conformal parametrization*

$$F : \mathbb{C} \rightarrow M.$$

Idea of proof. Since the projection $M \rightarrow \mathbb{R}^2$ is a global diffeomorphism, the surface M is simply connected. By uniformization, its conformal type is either $\mathbb{C}$ or the disk $\mathbb{D}$. In conformal coordinates, the coordinate functions of a minimal immersion are harmonic; in particular, the projection $(x_1, x_2) : M \rightarrow \mathbb{R}^2$ becomes a harmonic diffeomorphism from the conformal parameter domain onto $\mathbb{R}^2$. A theorem of Heinz rules out a harmonic diffeomorphism from the disk onto the whole plane. Therefore the conformal type is $\mathbb{C}$. This is the standard Heinz–Osserman conformal-type lemma; see [5, Chapter 5]. $\square$

Lemma 5.5 (Holomorphicity of the Gauss map). *Let $F : \Sigma \rightarrow \mathbb{R}^3$ be a conformal minimal immersion. Its oriented unit normal map*

$$\nu : \Sigma \rightarrow \mathbb{S}^2$$

has meromorphic stereographic projection. More precisely, if

$$g = \frac{\nu_1 + i\nu_2}{1 + \nu_3},$$

where this expression is finite, then g is holomorphic.

Sketch. In a conformal coordinate z , minimality is equivalent to harmonicity of the coordinate functions of F , so

$$\Phi := 2\partial_z F$$

is a holomorphic $\mathbb{C}^3$ -valued function. Conformality gives the null condition

$$\Phi_1^2 + \Phi_2^2 + \Phi_3^2 = 0.$$

The Gauss map can be recovered from the complex tangent line spanned by Φ , and in the usual Weierstrass notation one may write

$$g = \frac{\Phi_3}{\Phi_1 - i\Phi_2}$$

up to the harmless choice of stereographic pole. Hence g is meromorphic. If the normal avoids the stereographic pole, then g is holomorphic. $\square$

Proof of Theorem 5.3. Let

$$M = \text{graph } u \subset \mathbb{R}^3$$

and orient M by the upward unit normal

$$\nu = \frac{(-u_x, -u_y, 1)}{\sqrt{1 + |Du|^2}}.$$

Then

$$\nu_3 = \frac{1}{\sqrt{1 + |Du|^2}} > 0,$$

so the Gauss map of M takes values in the open upper hemisphere.

By Lemma 5.4, choose a global conformal parametrization $F : \mathbb{C} \rightarrow M$. By Lemma 5.5, the stereographic Gauss map

$$g(z) = \frac{\nu_1(z) + i\nu_2(z)}{1 + \nu_3(z)}$$

is holomorphic on $\mathbb{C}$. Since $\nu_3 > 0$,

$$|g|^2 = \frac{\nu_1^2 + \nu_2^2}{(1 + \nu_3)^2} = \frac{1 - \nu_3^2}{(1 + \nu_3)^2} = \frac{1 - \nu_3}{1 + \nu_3} < 1.$$

Thus g is a bounded entire holomorphic function. By Liouville's theorem, g is constant. Hence the unit normal ν is constant on M .

If the unit normal is constant, then the tangent plane is constant. Equivalently, for the constant vector $\nu_0 = \nu$,

$$\nabla_M(x \cdot \nu_0) = 0,$$

so $x \cdot \nu_0$ is constant on the connected surface M . Therefore M is contained in an affine plane. Since M is a graph over the horizontal plane, this affine plane is not vertical, and so

$$u(x, y) = ax + by + c$$

for constants a, b, c . This proves the theorem. $\square$

Remark 5.6. The proof uses the global graph condition in two places. First, the upward normal stays in one open hemisphere, so the Gauss map becomes a bounded holomorphic function. Second, the graph is entire, so its conformal type is $\mathbb{C}$. Without either global input, minimal surfaces in $\mathbb{R}^3$ can be highly non-flat; the catenoid and helicoid are the basic reminders.

5.3. Higher-Dimensional Bernstein Theorems. The classical theorem has a sharp higher-dimensional analogue. In the scalar graphical case the answer is positive up to dimension 7 and false starting in dimension 8.

Theorem 5.7 (Higher-dimensional Bernstein theorem). *Let $u : \mathbb{R}^n \rightarrow \mathbb{R}$ be a smooth entire solution of the minimal surface equation*

$$\operatorname{div} \left(\frac{Du}{\sqrt{1 + |Du|^2}} \right) = 0.$$

If $n \leq 7$, then u is affine. If $n \geq 8$, there exist non-affine smooth entire solutions.

Historically, the theorem was proved in stages:

domain dimension	result	main name
$n = 2$	true	Bernstein
$n = 3$	true	De Giorgi
$n = 4$	true	Almgren
$n \leq 7$	true	Simons
$n \geq 8$	false	Bombieri–De Giorgi–Giusti

The sharp counterexamples in dimension 8 and higher are due to Bombieri–De Giorgi–Giusti [3]. Their construction is organized by the Simons cone

$$C_S = \{(x, y) \in \mathbb{R}^4 \times \mathbb{R}^4 : |x| = |y|\} \subset \mathbb{R}^8,$$

which is an area-minimizing hypersurface cone. Very roughly, the two regions $|x| > |y|$ and $|x| < |y|$ provide the geometry that allows one to build a non-affine entire minimal graph in one higher dimension.

Remark 5.8 (Why the number 7 appears). The number 7 is not accidental. The graph of a scalar solution is an n -dimensional hypersurface in $\mathbb{R}^{n+1}$ and is locally mass-minimizing by the calibration argument from the previous subsection. If one studies such a graph at larger and larger scales, one is led to minimizing cones.

The codimension-one regularity theory developed earlier says that minimizing hypersurface cones are forced to be planes in dimensions at most 6, while the Simons cone gives the first genuine singular minimizing cone, of dimension 7. The Bernstein theorem still remains true for $n = 7$ because the cone analysis is combined with the extra global graphical structure; the failure in $n = 8$ is built from the 7-dimensional Simons cone.

Thus the Bernstein theorem and the regularity theorem for mass-minimizing hypersurfaces are two faces of the same cone-classification problem, with a one-dimensional shift between the singular cone and the graph counterexample.

Let us spell out the geometric mechanism behind the positive part of Theorem 5.7. For an entire solution u , set

$$M = \text{graph } u \subset \mathbb{R}^{n+1}, \quad T_u = \partial \llbracket E_u \rrbracket, \quad V_u = |T_u|.$$

As before, T_u is locally mass-minimizing and $\text{supp } T_u = M$. Consider the dilations

$$T_{u,r} := (\eta_{0,r})_{\#} T_u, \quad V_{u,r} := |T_{u,r}|, \quad r \rightarrow \infty.$$

After passing to a subsequence, compactness for minimizing currents gives a limiting minimizing cone T_∞ at infinity, and the associated varifolds satisfy

$$V_{u,r_i} \rightharpoonup |T_\infty|.$$

This cone should be thought of as the asymptotic shape of the graph.

If all possible such cones are multiplicity-one planes, then Allard regularity gives that $\text{supp } T_{u,r}$ is graphical over a plane with small $C^{1,\alpha}$ norm on fixed balls for large r . Scaling back, the original graph becomes flatter and flatter at infinity. The analytic part of the Bernstein theorem then upgrades this decay into a Liouville conclusion: the gradient is forced to be constant, hence u is affine.

The difficult point is therefore to exclude non-planar limiting behavior at infinity. In dimensions $n \leq 7$, the De Giorgi–Almgren–Simons argument combines cone analysis, stability, and global graphical estimates to force flatness. In dimension 8, the Simons cone is available as a stable area-minimizing model in the base geometry, and Bombieri–De Giorgi–Giusti showed that it can be used to construct a non-affine entire minimal graph.

This explains the sharp dichotomy in Theorem 5.7.

Remark 5.9 (A useful way to remember the theorem). For scalar entire minimal graphs,

$$\boxed{n \leq 7 \implies \text{rigidity}, \quad n \geq 8 \implies \text{counterexamples}}.$$

For vector-valued graphs the story is different: Lipschitz regularity no longer forces smoothness in the same way, as seen earlier from the Lawson–Osseman type example involving the Hopf fibration. Thus the Bernstein theorem above is genuinely a codimension-one scalar phenomenon.

5.4. Bounded Slope and Moser’s Theorem. The previous theorem says that high-dimensional entire minimal graphs need not be planes. There is, however, a dimension-free rigidity theorem if one assumes a global slope bound. We keep the same notation as above: for a scalar entire graph $M = \text{graph } u$,

$$E_u = \{(x, t) : t < u(x)\}, \quad T_u = \partial \llbracket E_u \rrbracket, \quad V_u = |T_u|.$$

Theorem 5.10 (Moser’s bounded-slope Bernstein theorem). *Let $u : \mathbb{R}^n \rightarrow \mathbb{R}$ be a smooth entire solution of (MSE). If*

$$\sup_{\mathbb{R}^n} |Du| < \infty,$$

then u is affine.

Proof. Write the minimal surface equation as

$$D_i A_i(Du) = 0, \quad A(p) = \frac{p}{\sqrt{1 + |p|^2}}.$$

Fix $k \in \{1, \dots, n\}$ and set $w_k = D_k u$. Differentiating the equation gives

$$D_i (a^{ij}(Du) D_j w_k) = 0,$$

where

$$a^{ij}(p) = \frac{\partial A_i}{\partial p_j}(p) = \frac{\delta_{ij}}{\sqrt{1+|p|^2}} - \frac{p_i p_j}{(1+|p|^2)^{\frac{3}{2}}}.$$

If $|Du| \leq L$, then this linear equation is uniformly elliptic. Indeed, for every $\xi \in \mathbb{R}^n$,

$$a^{ij}(Du)\xi_i\xi_j = \frac{(1+|Du|^2)|\xi|^2 - (Du \cdot \xi)^2}{(1+|Du|^2)^{\frac{3}{2}}},$$

and therefore

$$\frac{1}{(1+L^2)^{\frac{3}{2}}}|\xi|^2 \leq a^{ij}(Du)\xi_i\xi_j \leq |\xi|^2.$$

By Moser's Harnack inequality, equivalently the De Giorgi–Moser oscillation estimate for uniformly elliptic divergence-form equations [10], there are constants $C < \infty$ and $\alpha \in (0, 1)$, depending only on n and L , such that for $0 < \rho < R$,

$$\operatorname{osc}_{B_\rho(0)} w_k \leq C \left(\frac{\rho}{R} \right)^\alpha \operatorname{osc}_{B_R(0)} w_k.$$

Since $|w_k| \leq L$ on all of $\mathbb{R}^n$, we get

$$\operatorname{osc}_{B_\rho(0)} w_k \leq 2CL \left(\frac{\rho}{R} \right)^\alpha.$$

Letting $R \rightarrow \infty$ gives $\operatorname{osc}_{B_\rho(0)} w_k = 0$. As ρ is arbitrary, $D_k u$ is constant on $\mathbb{R}^n$. This holds for every k , hence Du is constant and u is affine. $\square$

Remark 5.11 (Relation with the blow-down notation). The slope condition is invariant under the same rescaling used for tangent cones. If

$$u_r(x) := \frac{u(rx)}{r},$$

then u_r also solves (MSE), $|Du_r(x)| = |Du(rx)| \leq L$, and

$$E_{u_r} = \eta_{0,r}(E_u), \quad T_{u_r} = (\eta_{0,r})_\# T_u = T_{u,r}, \quad V_{u_r} = V_{u,r}.$$

Thus the analytic estimate above and the current compactness picture are looking at the same objects. Compactness says that T_{u,r_i} has subsequential minimizing conical limits; the bounded-slope theorem says that, under the global ellipticity supplied by $|Du| \leq L$, the only possible entire smooth graphical behavior is affine.

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